

Contribution Validation Report — ONE simulation run only

Case: all contributions demo

Duration: 600 s

Towers: 5

Receivers: 1

Max PR meas/epoch: 5

Max EKF rows/epoch: 5

Final pos error: 1137.4433 m

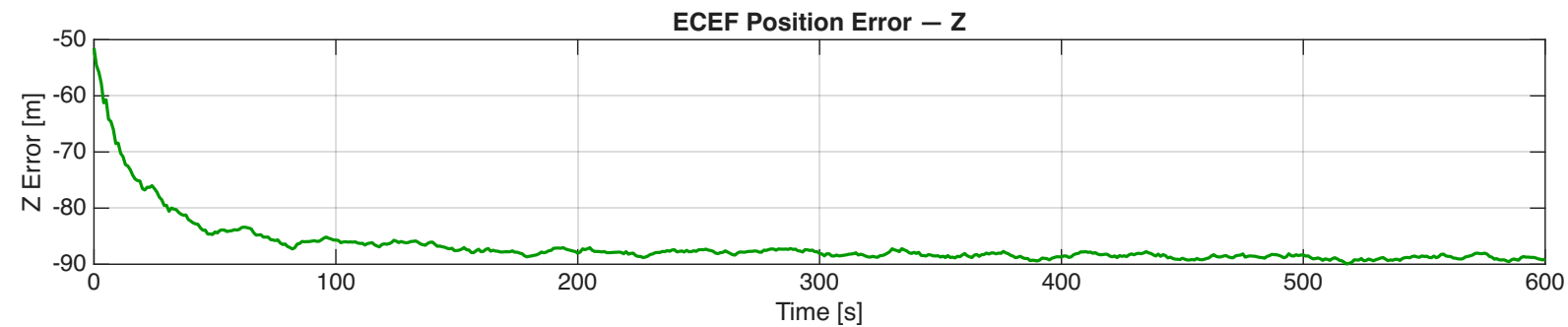
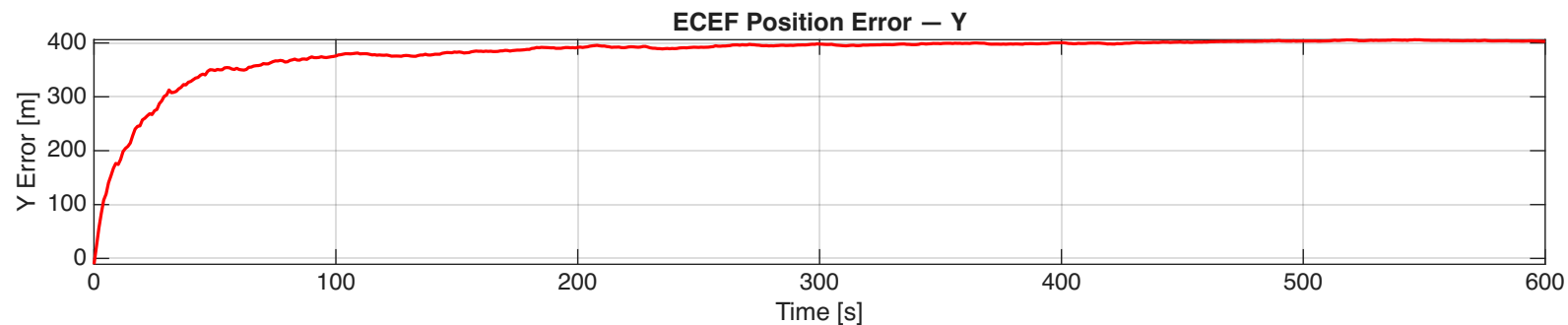
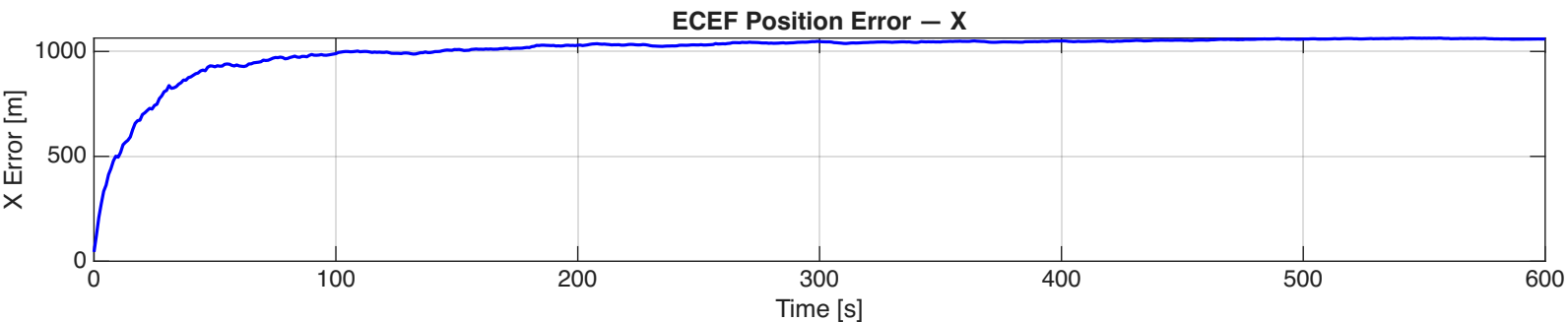
Pos RMS (last 20%): 1138.5228 m

Mean NIS: 5.27 (E[NIS] ~ 5)

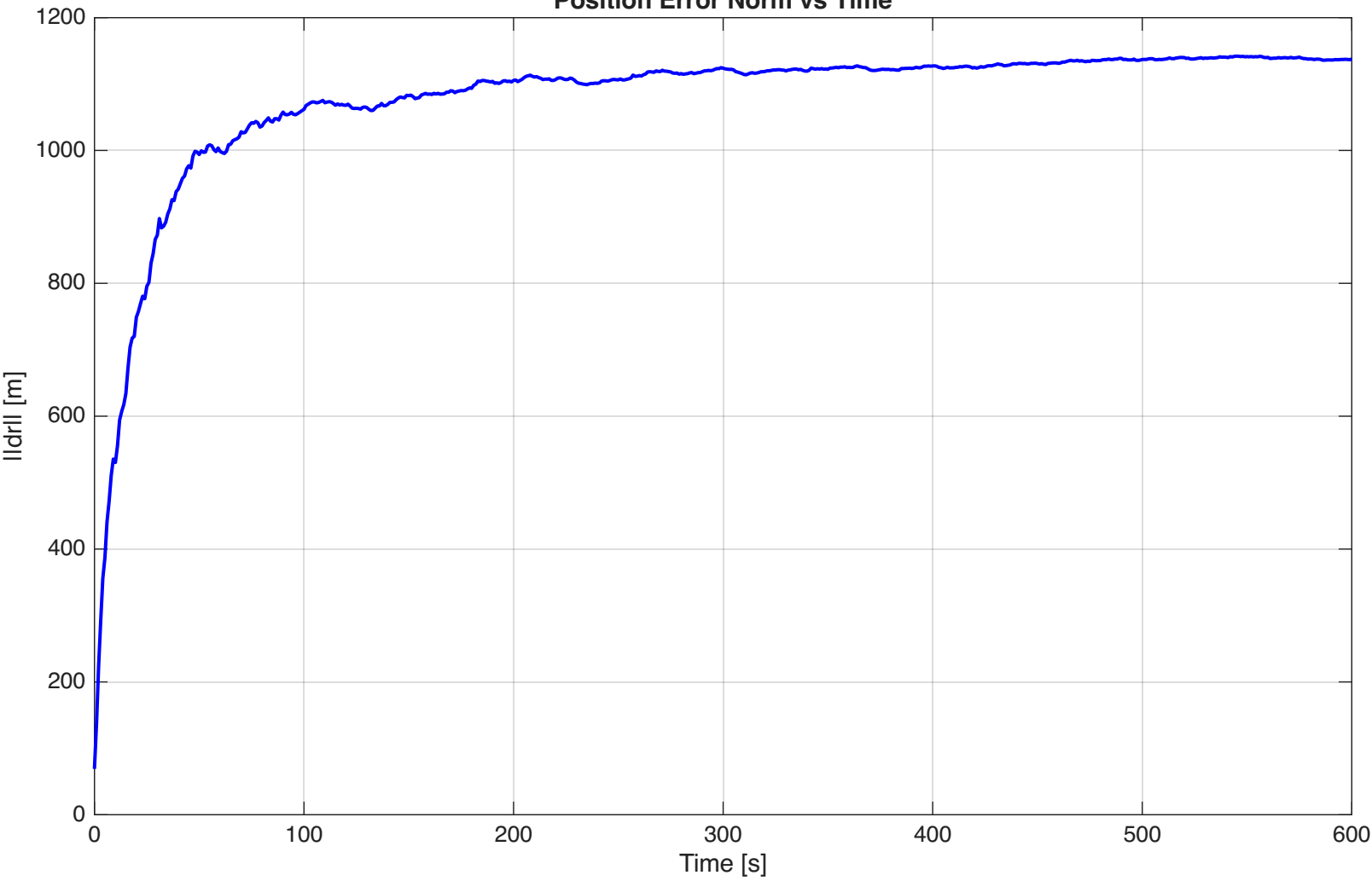
Enabled effects: Sagnac-truth, Sagnac-model, Shapiro-truth, TowerSurvey, PCO, PCV, CorrNoise, Trop, Iono, Doppler-diag, Carrier-diag

Output PDF: /Users/ludwigmatuschka/Library/CloudStorage/OneDrive-Persönlich/Dokumente/ICH/Karriere/Cranfield/IRP/Codes/IRP_Simulation/oo_v1/output/effect_contribution_validation_report.pdf

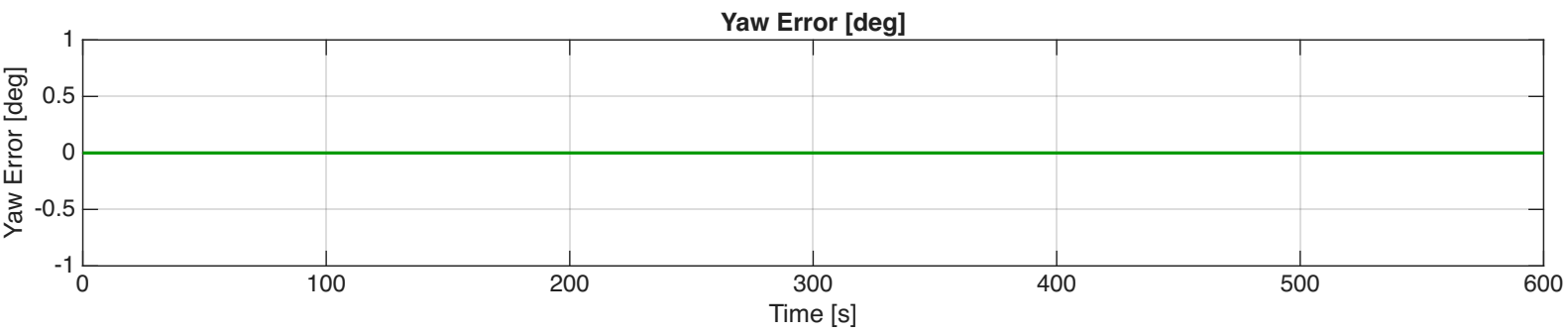
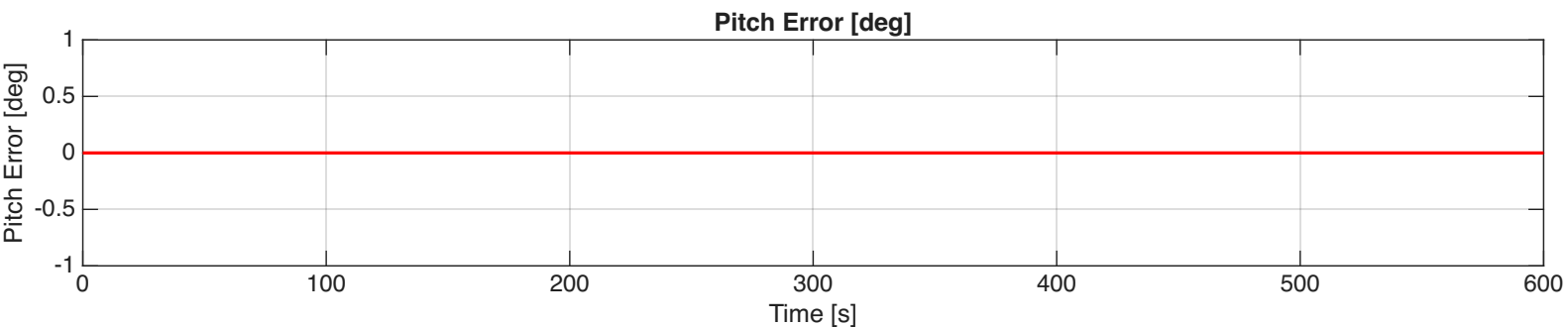
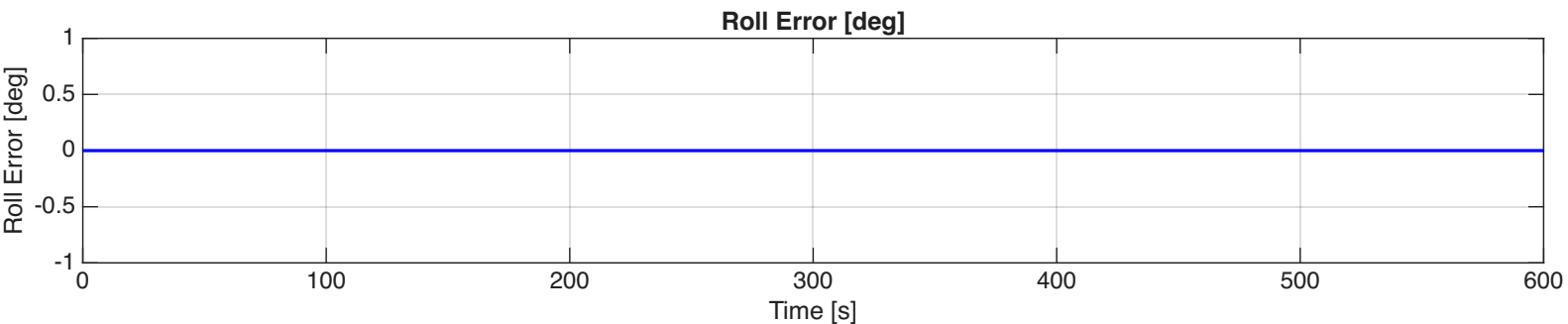
Position Error Components (ECEF)



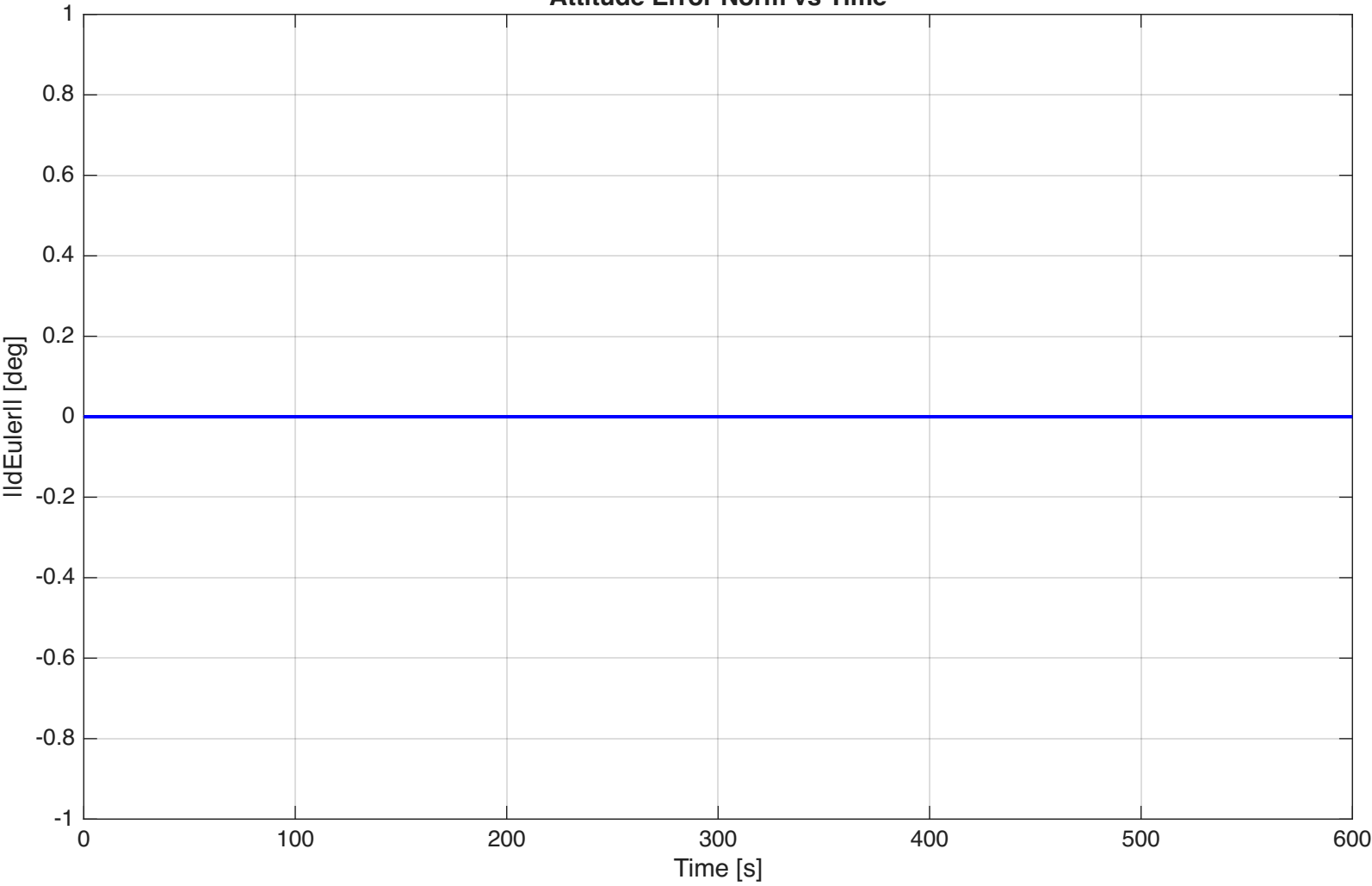
Position Error Norm vs Time



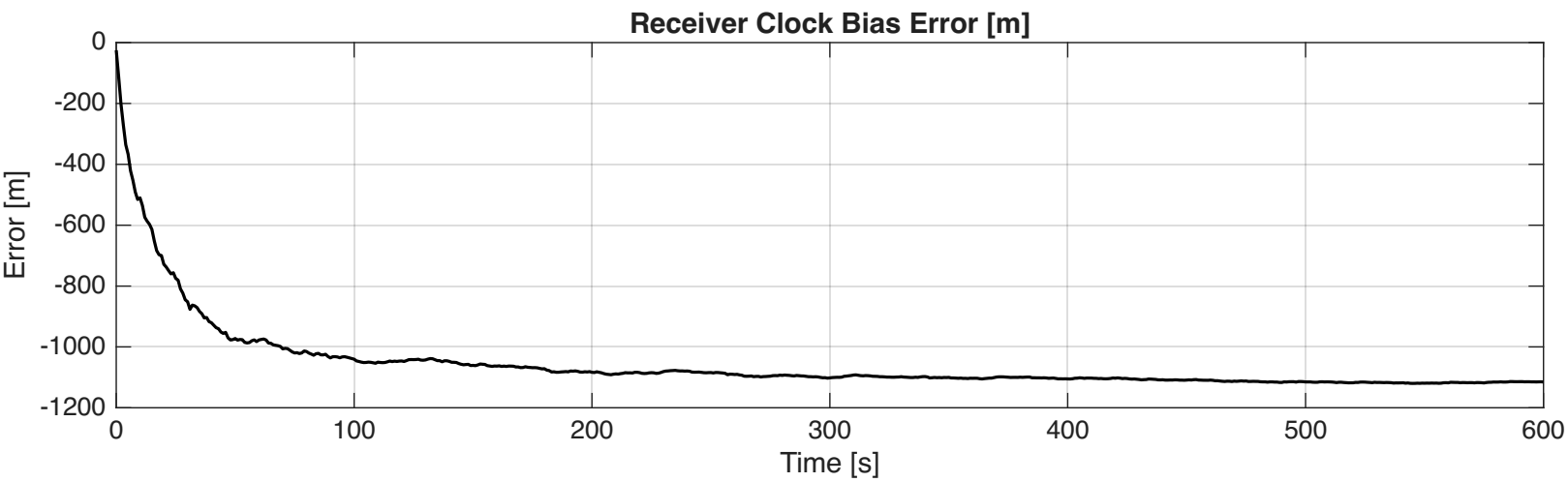
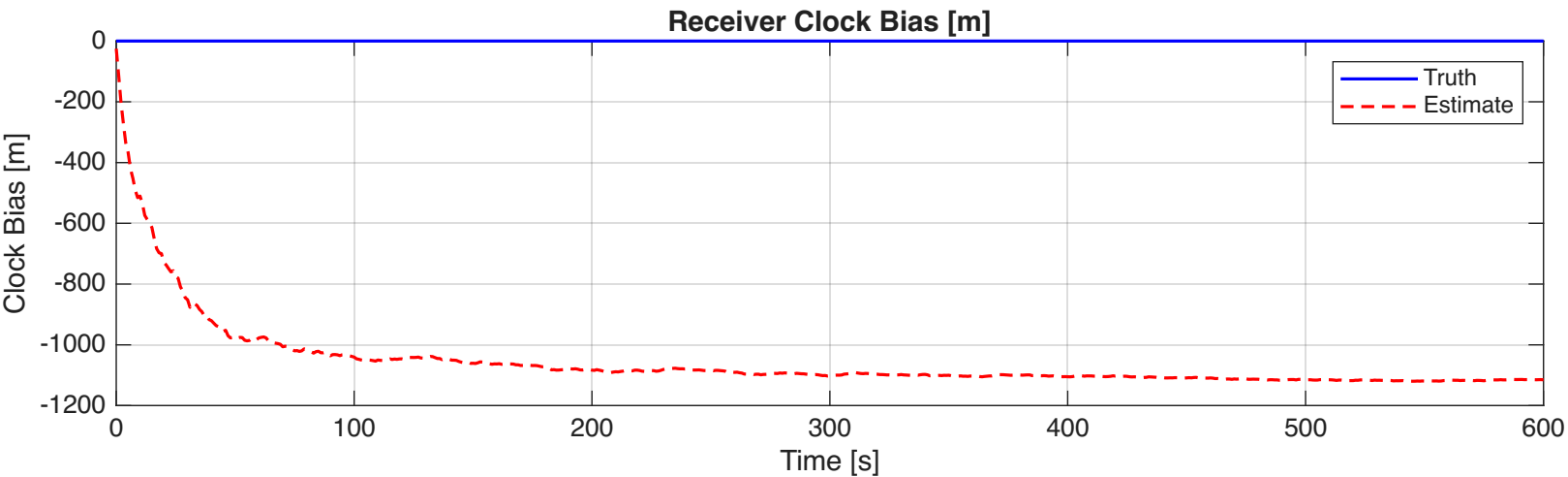
Attitude Error Components



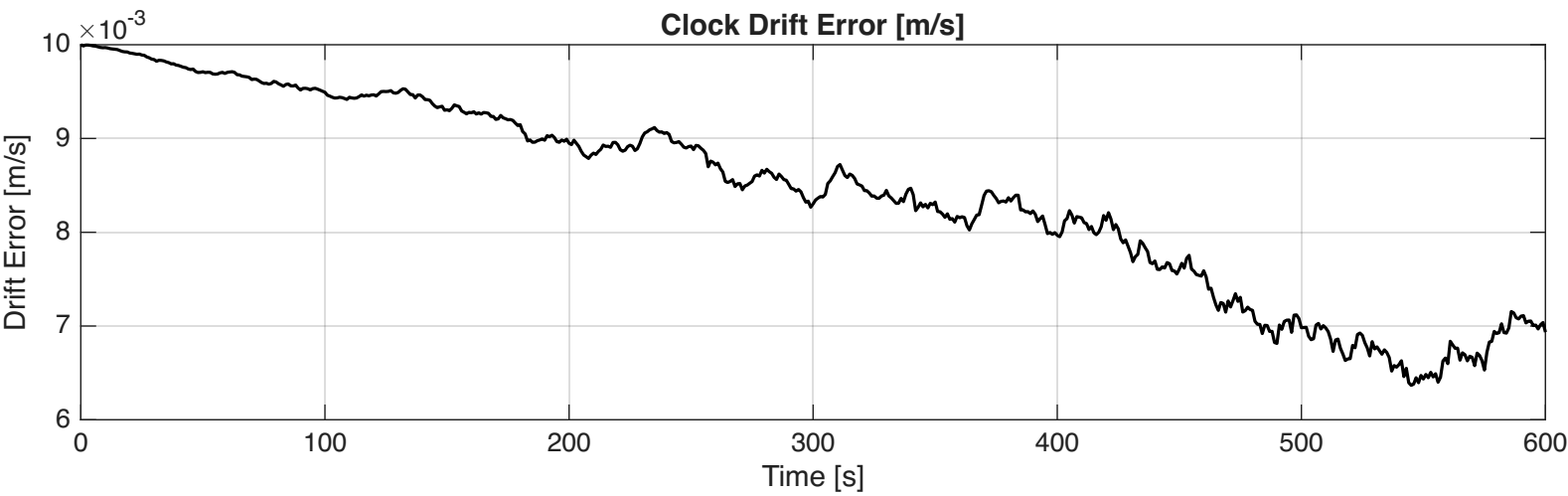
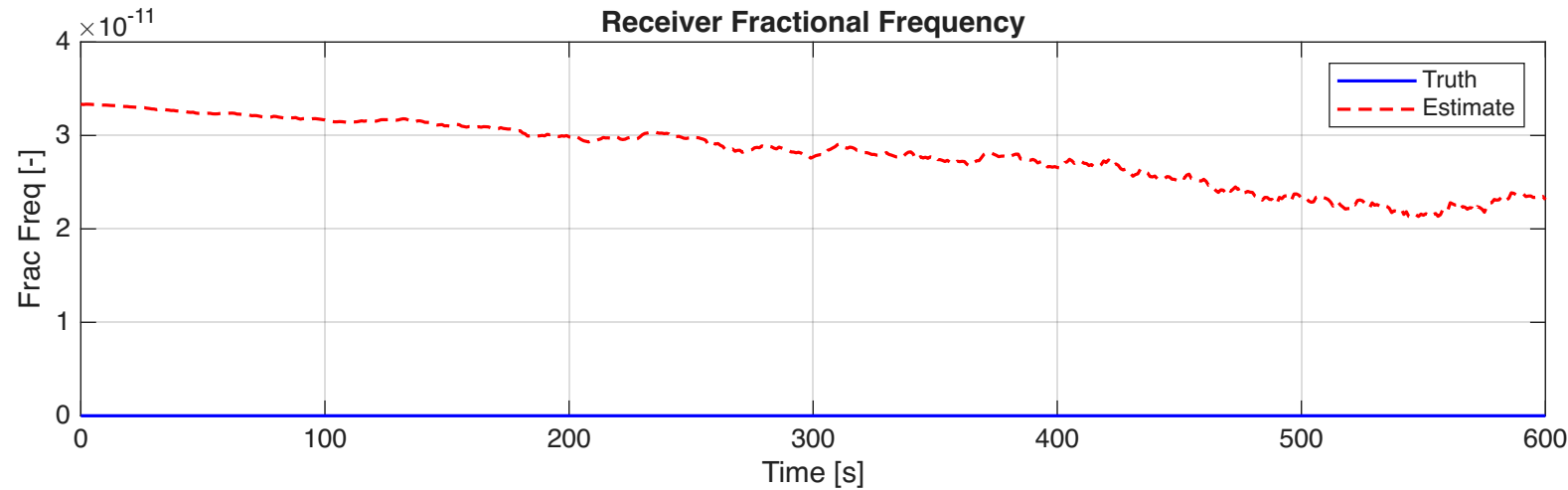
Attitude Error Norm vs Time



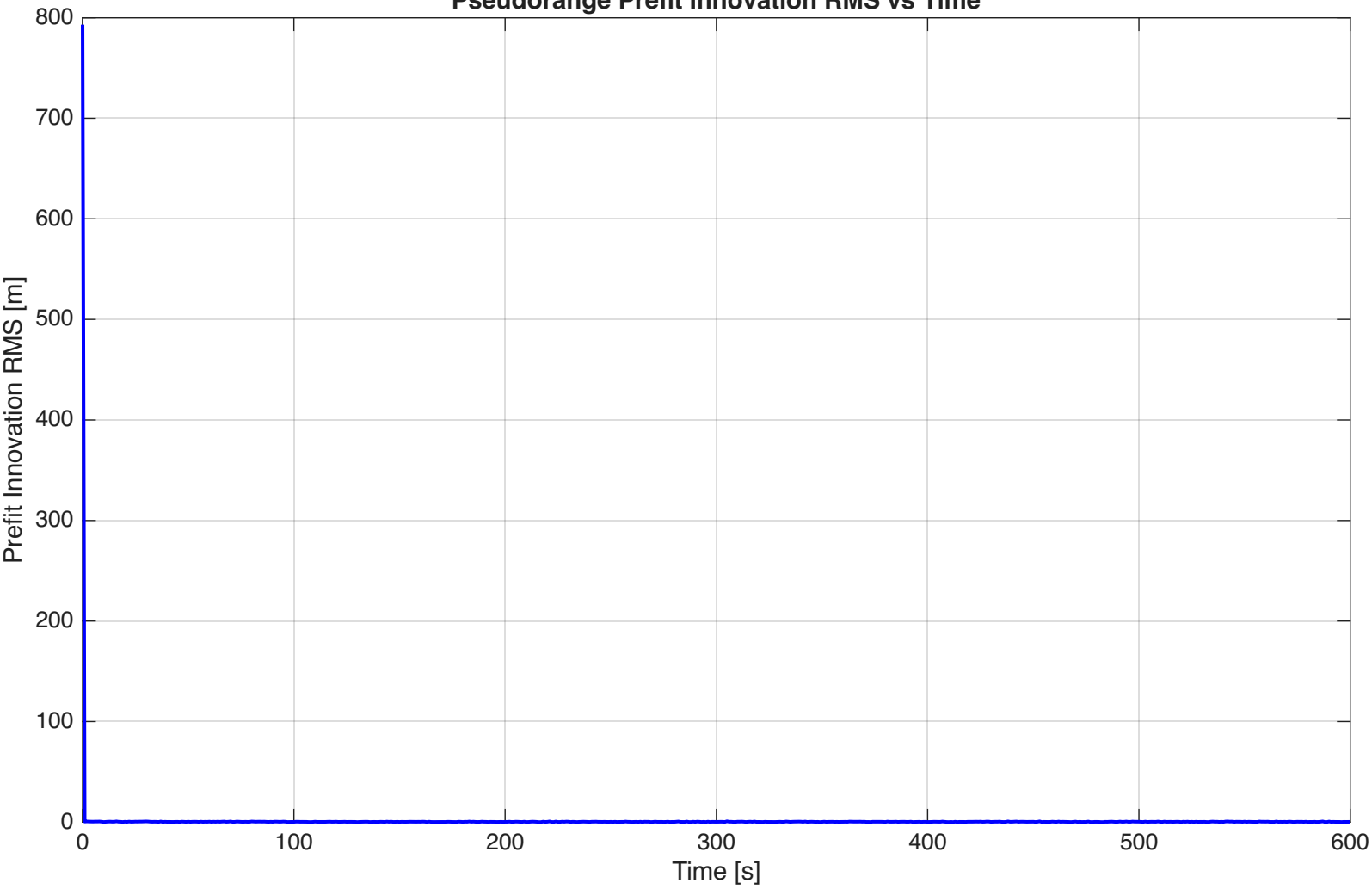
Receiver Clock Bias



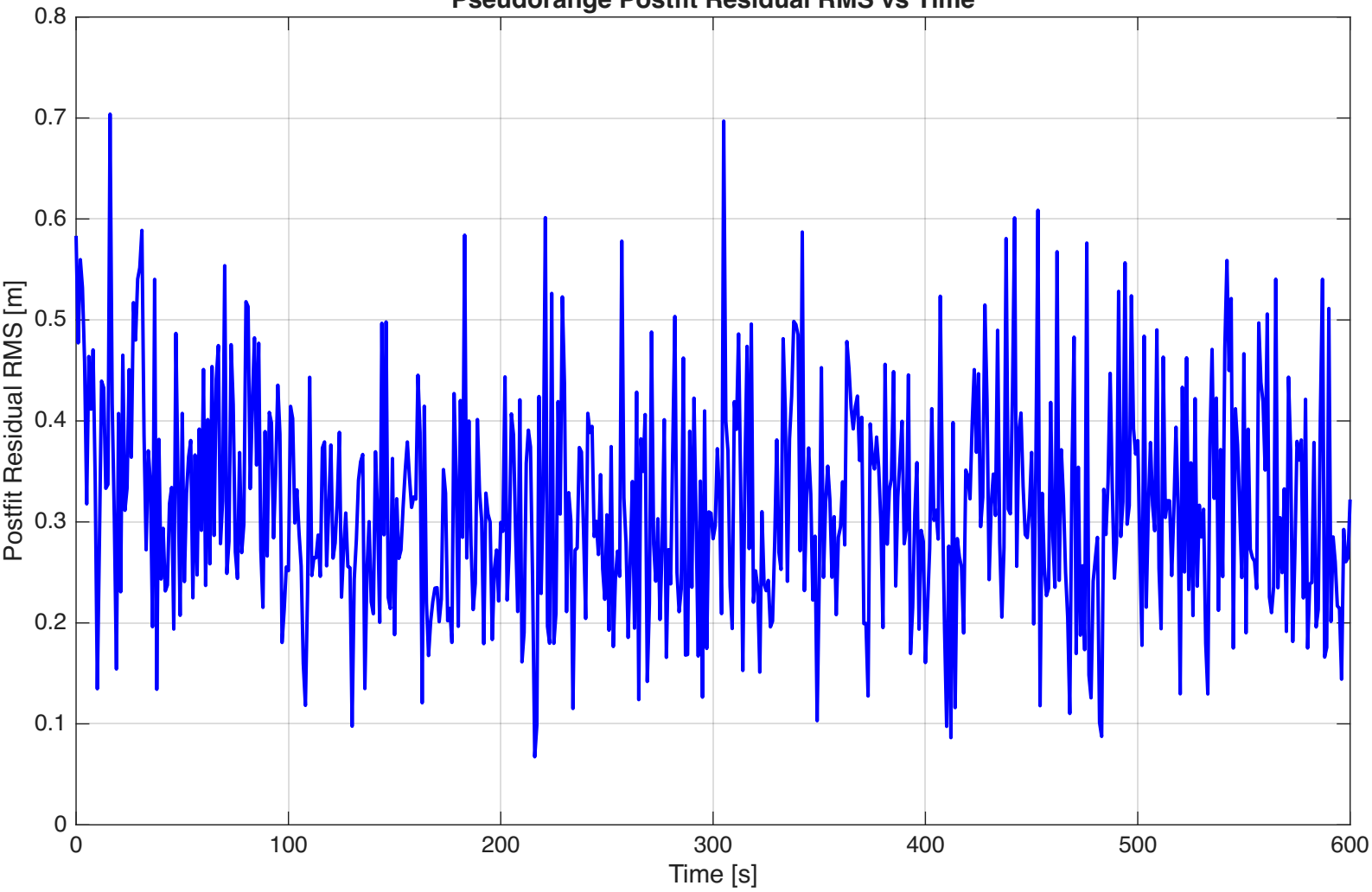
Receiver Clock Drift / Fractional Frequency



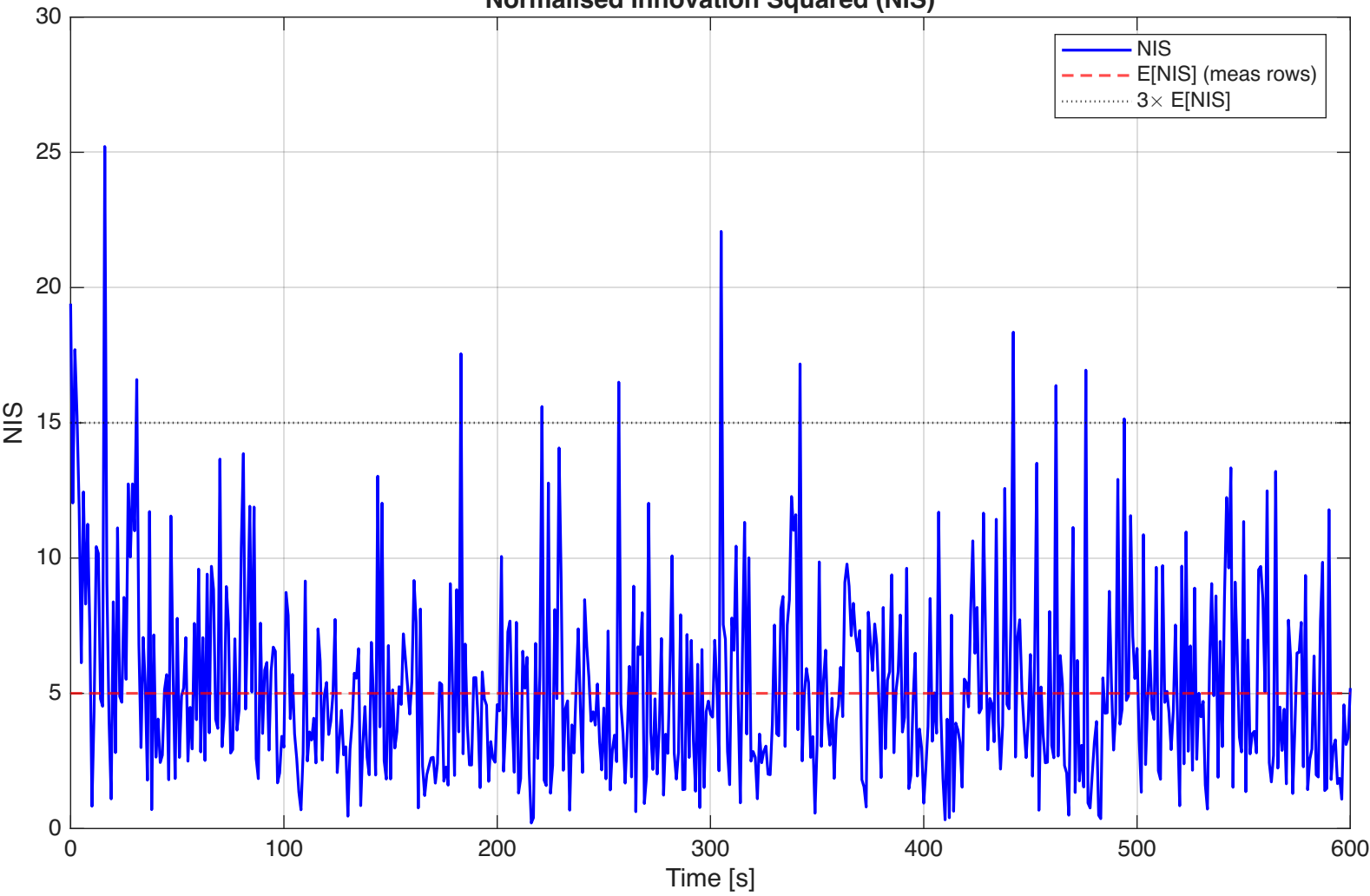
Pseudorange Prefit Innovation RMS vs Time



Pseudorange Postfit Residual RMS vs Time

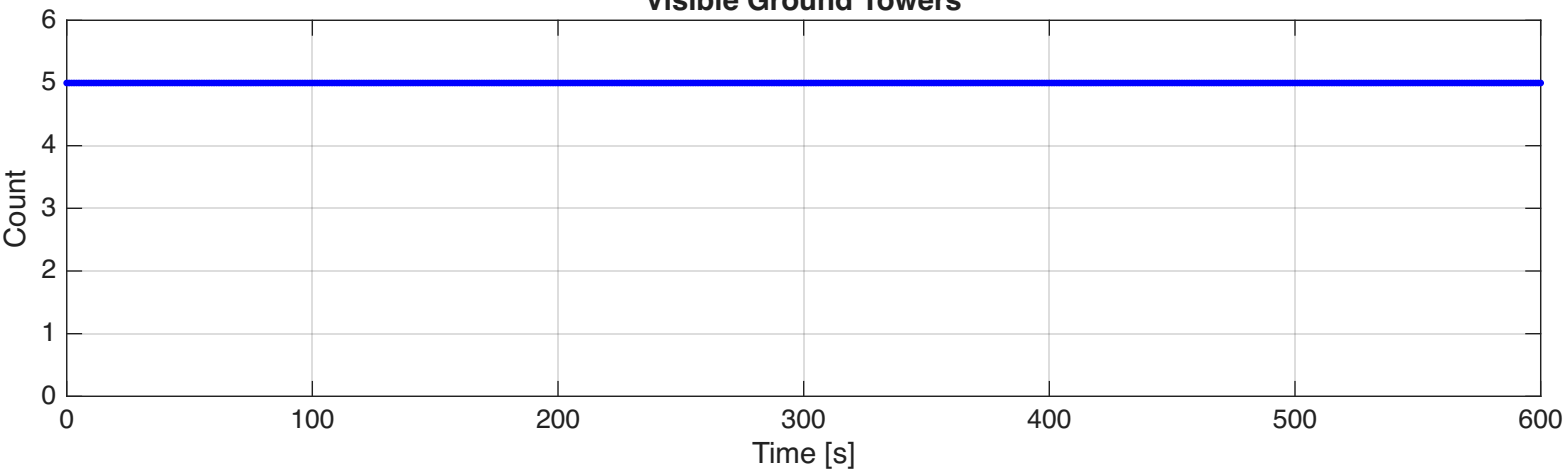


Normalised Innovation Squared (NIS)

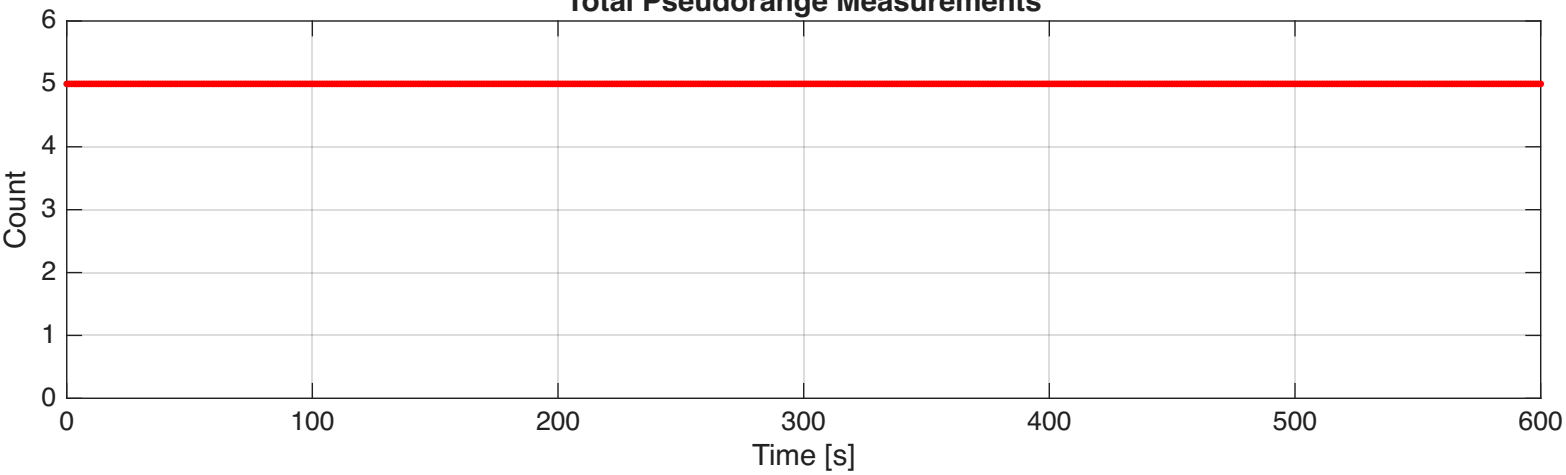


Observations per Epoch

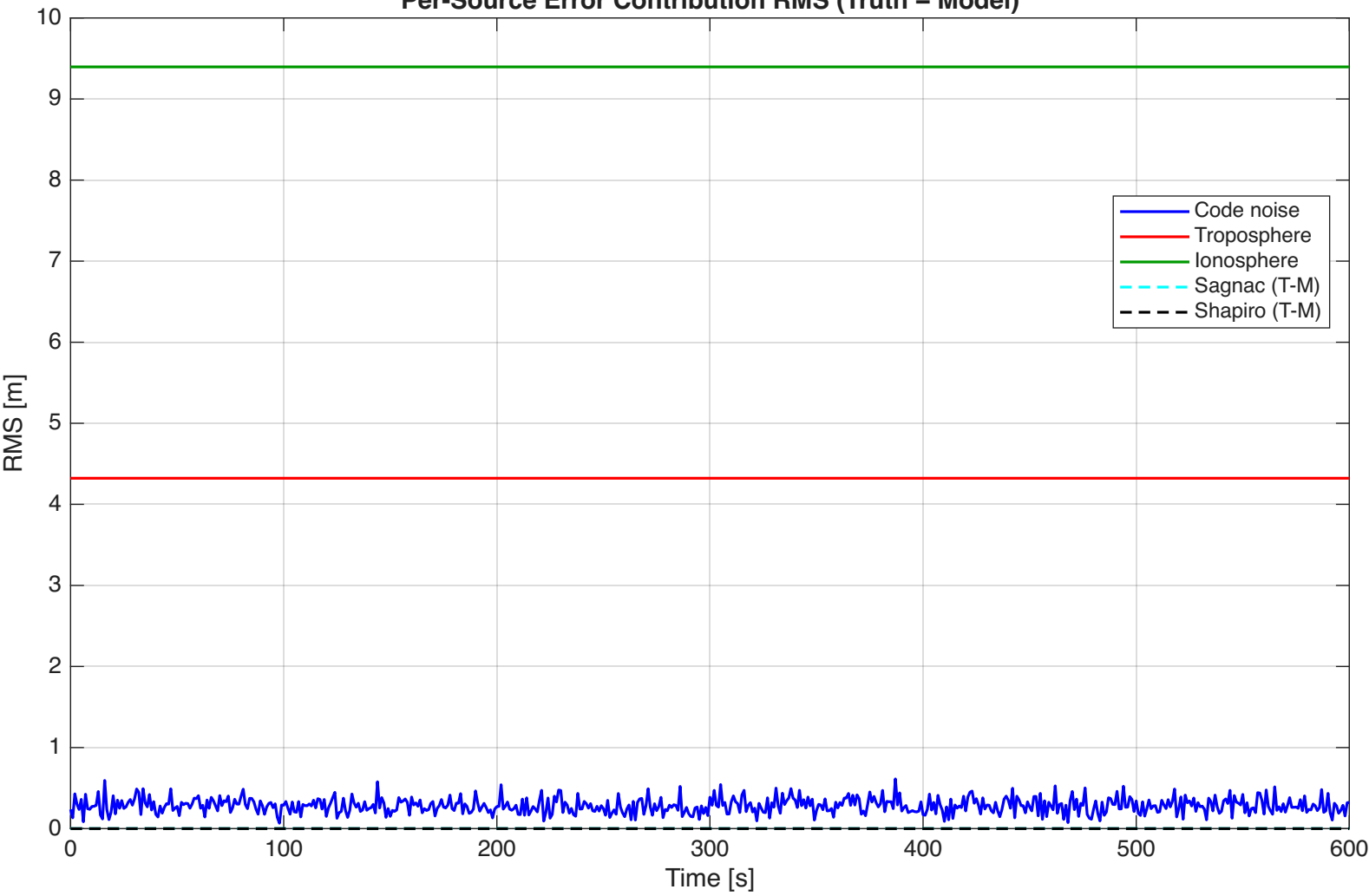
Visible Ground Towers

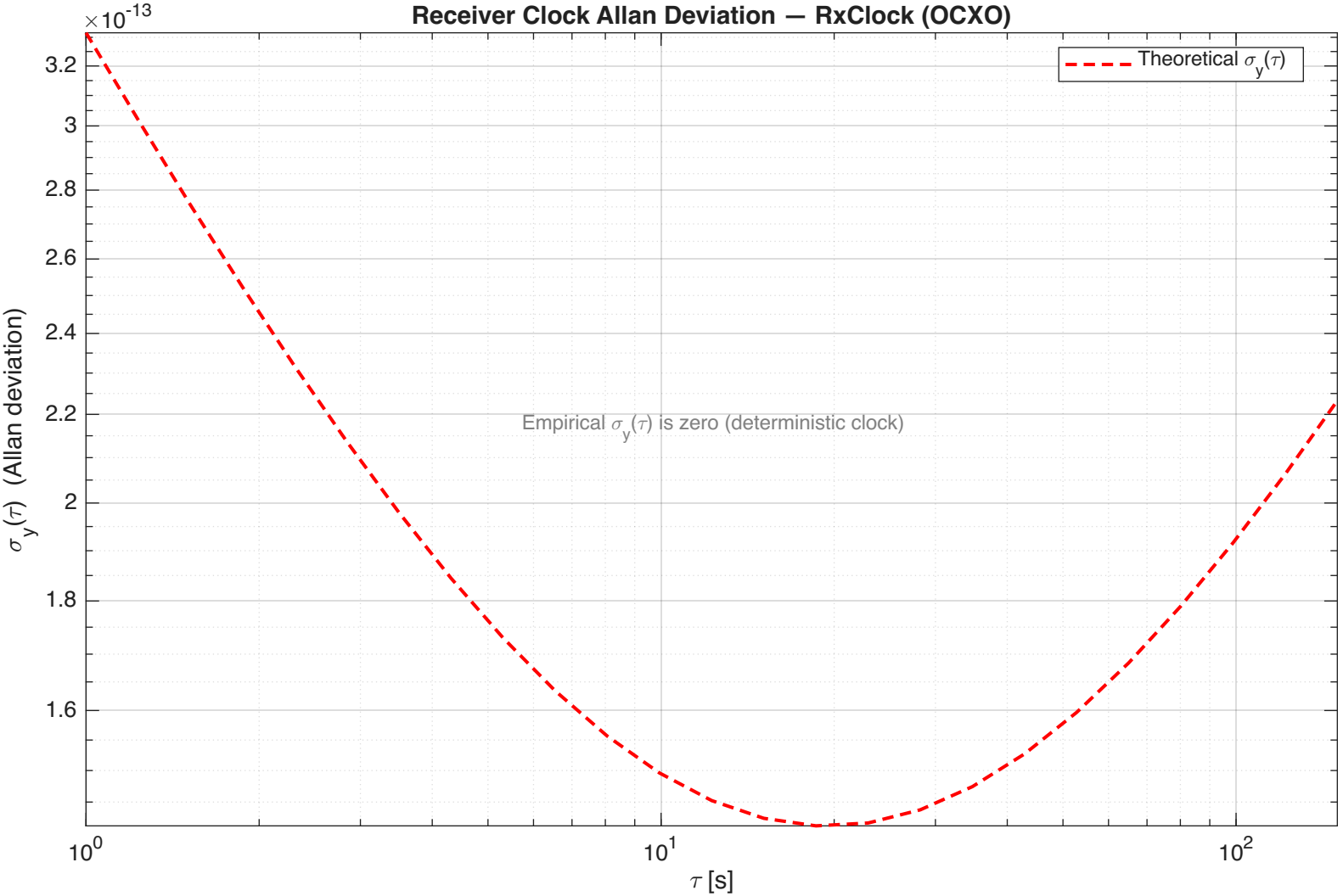


Total Pseudorange Measurements

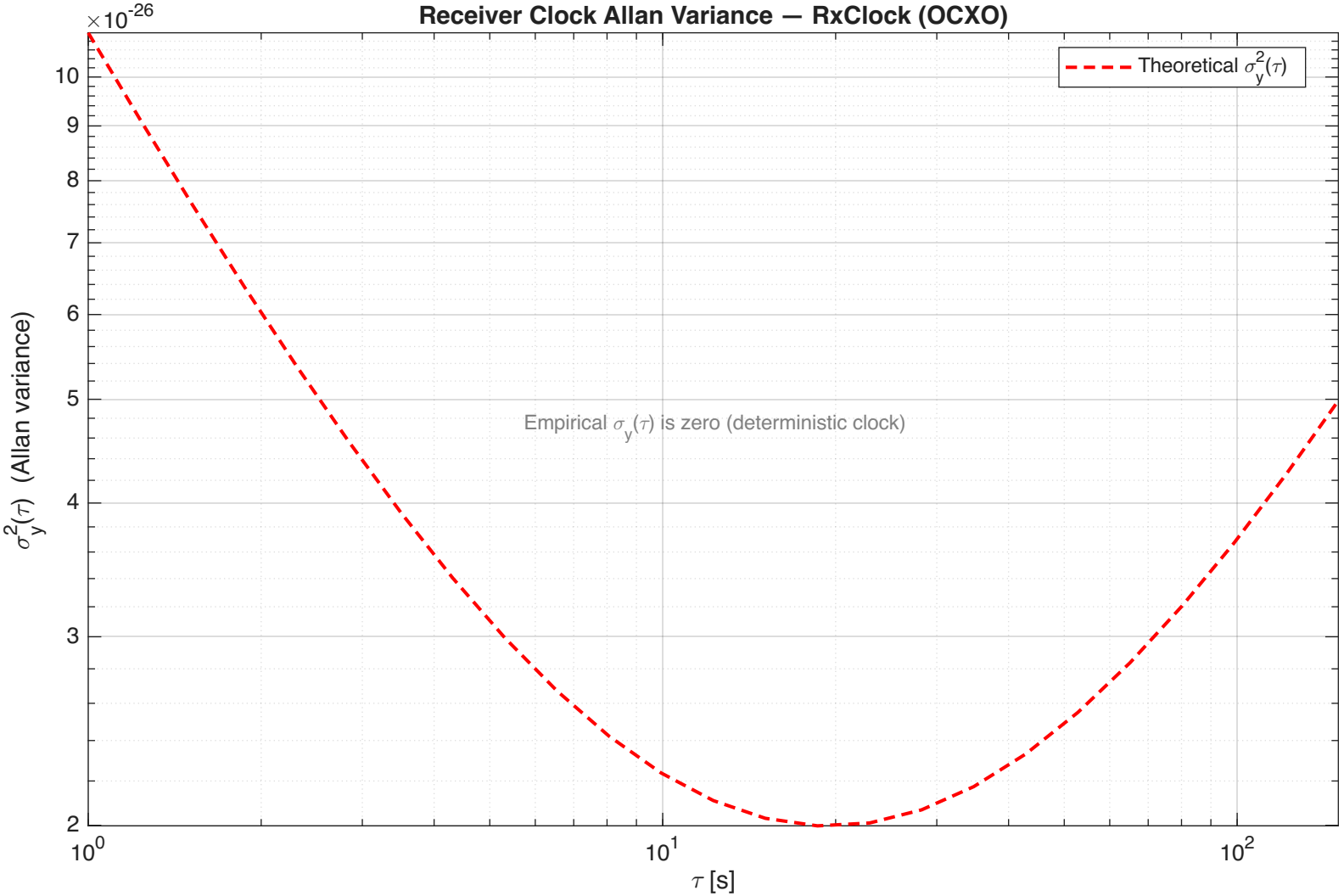


Per-Source Error Contribution RMS (Truth – Model)

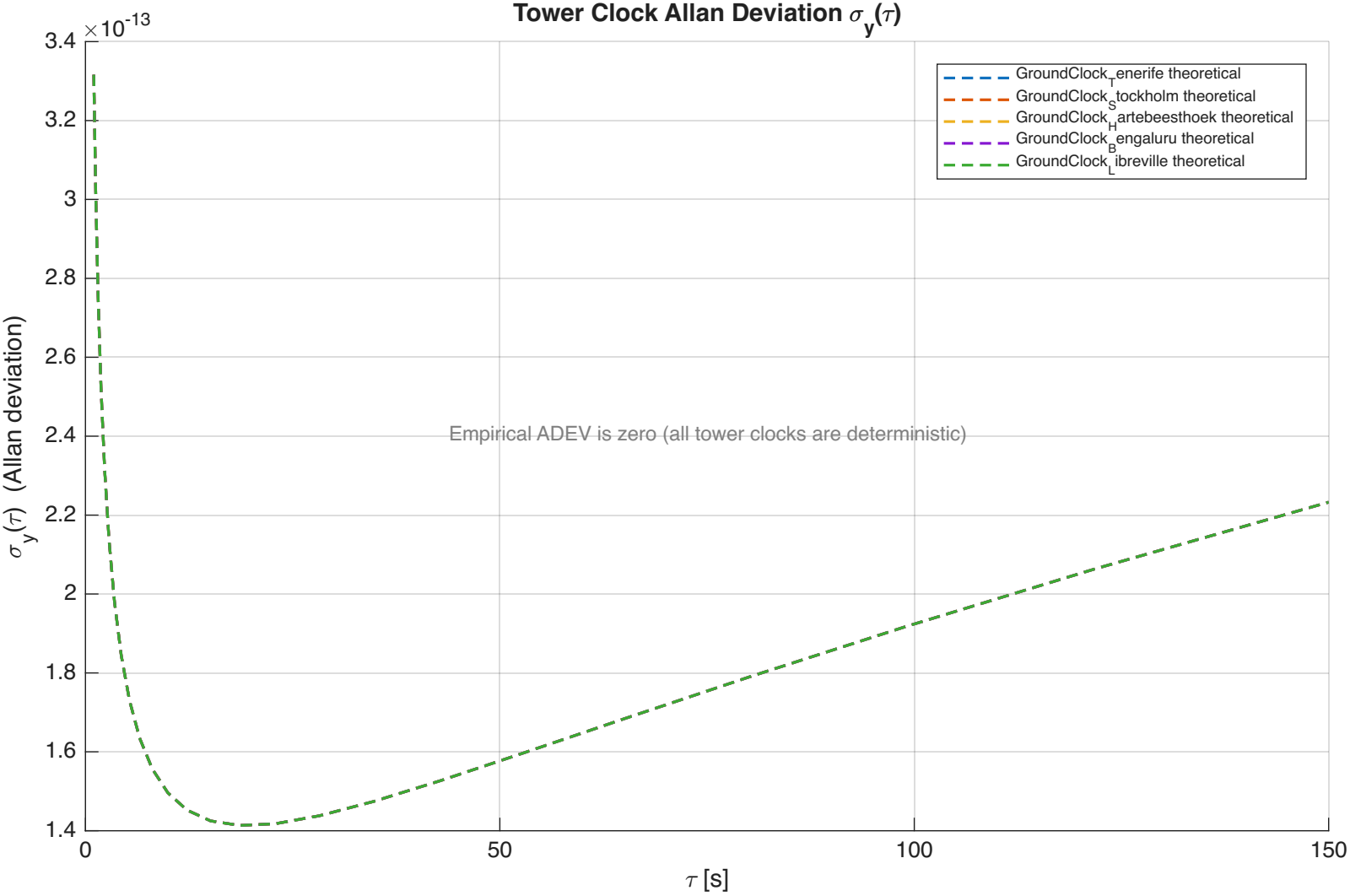




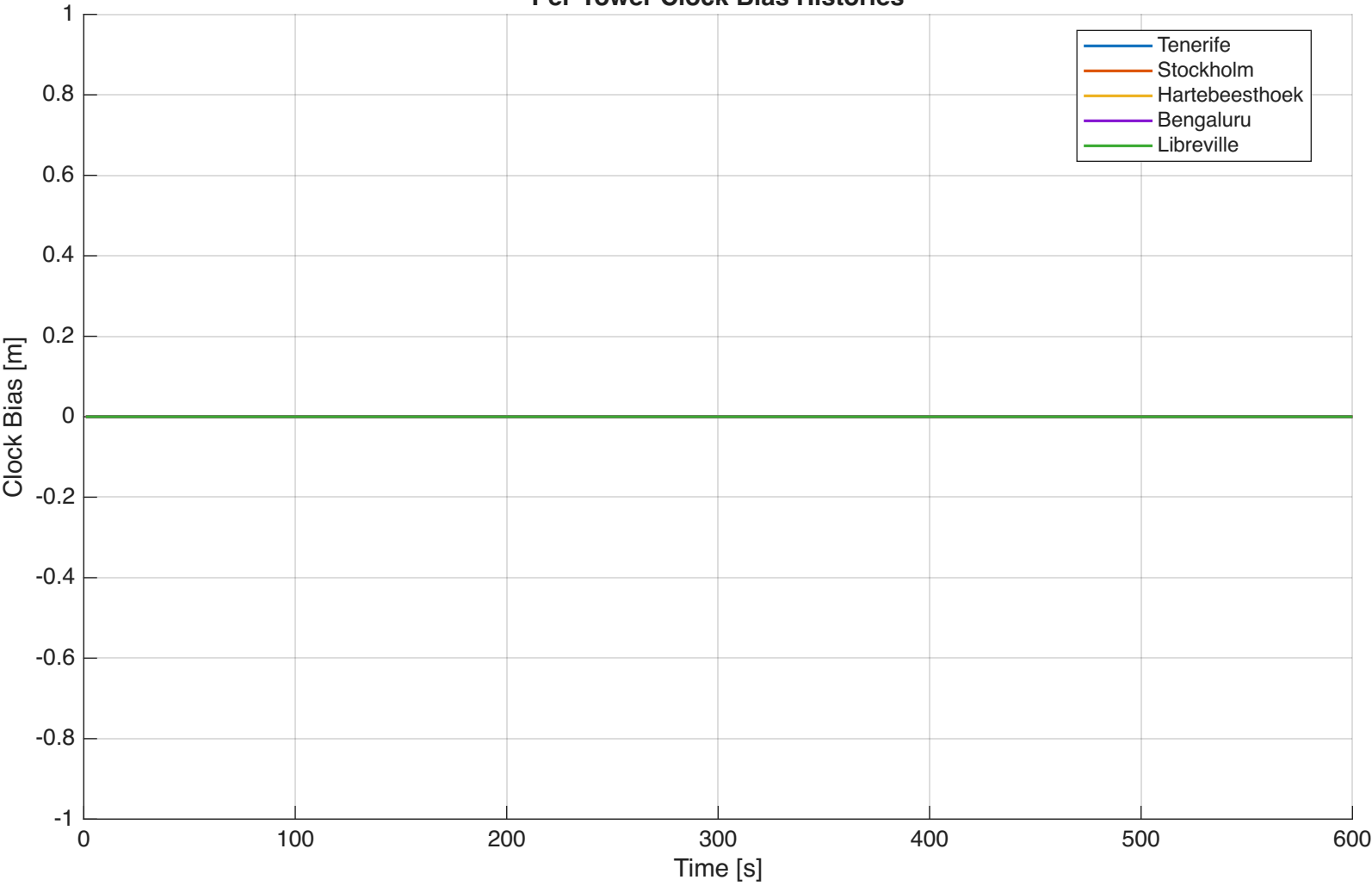
Receiver Clock Allan Variance — RxClock (OCXO)



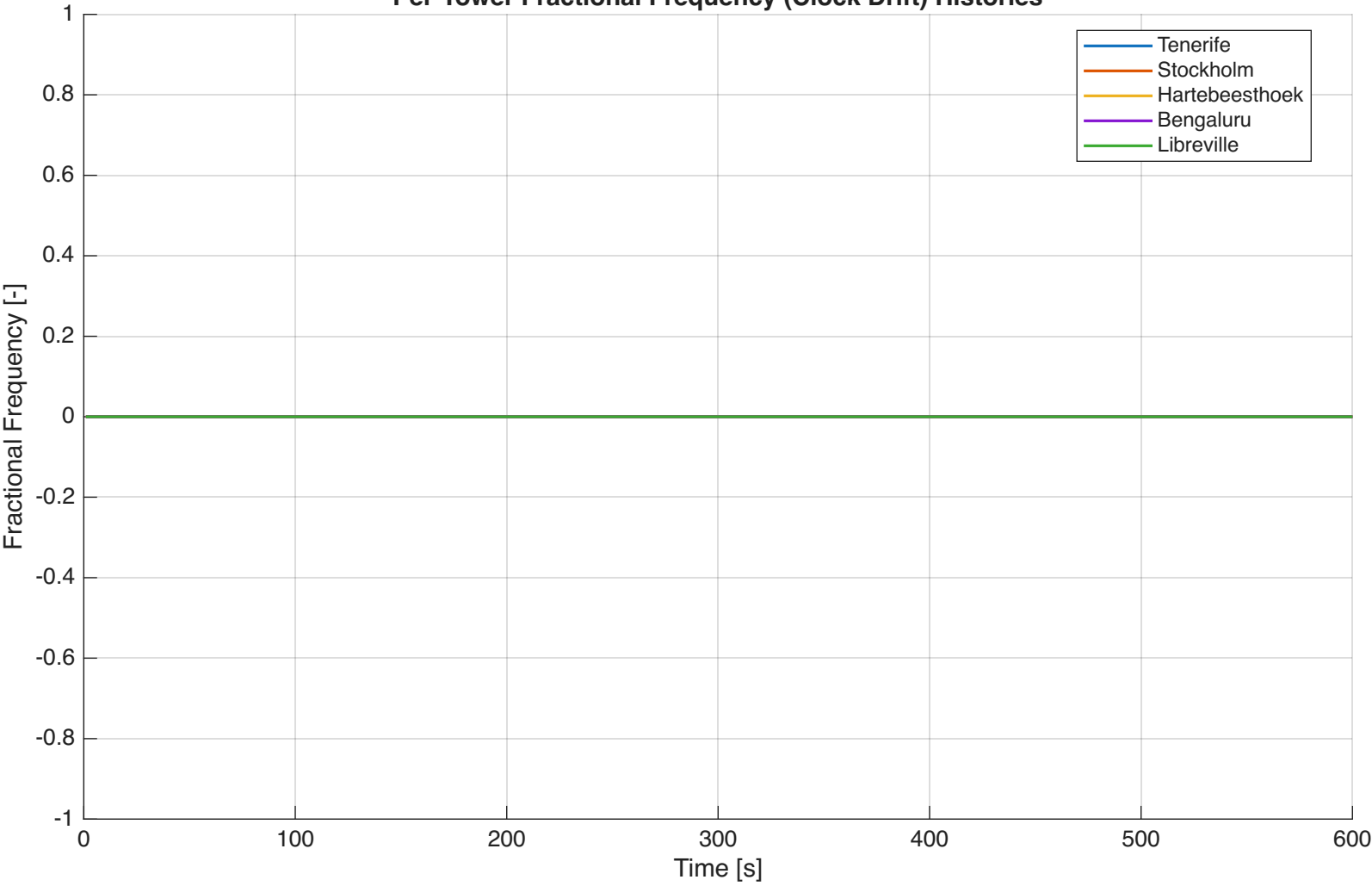
Tower Clock Allan Deviation $\sigma_y(\tau)$



Per-Tower Clock Bias Histories

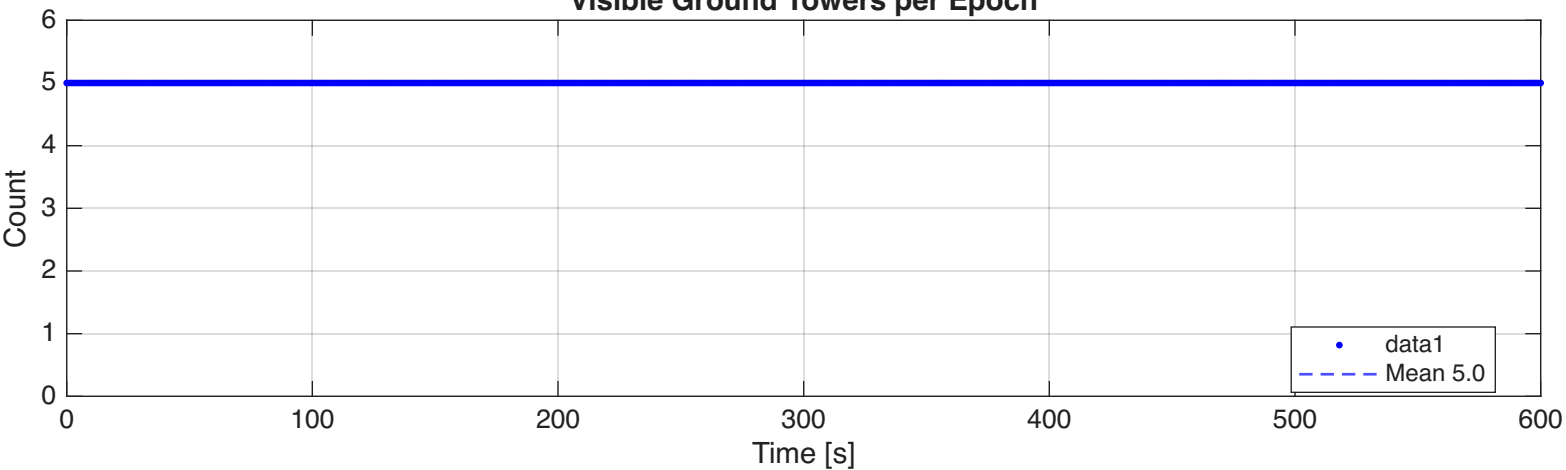


Per-Tower Fractional Frequency (Clock Drift) Histories

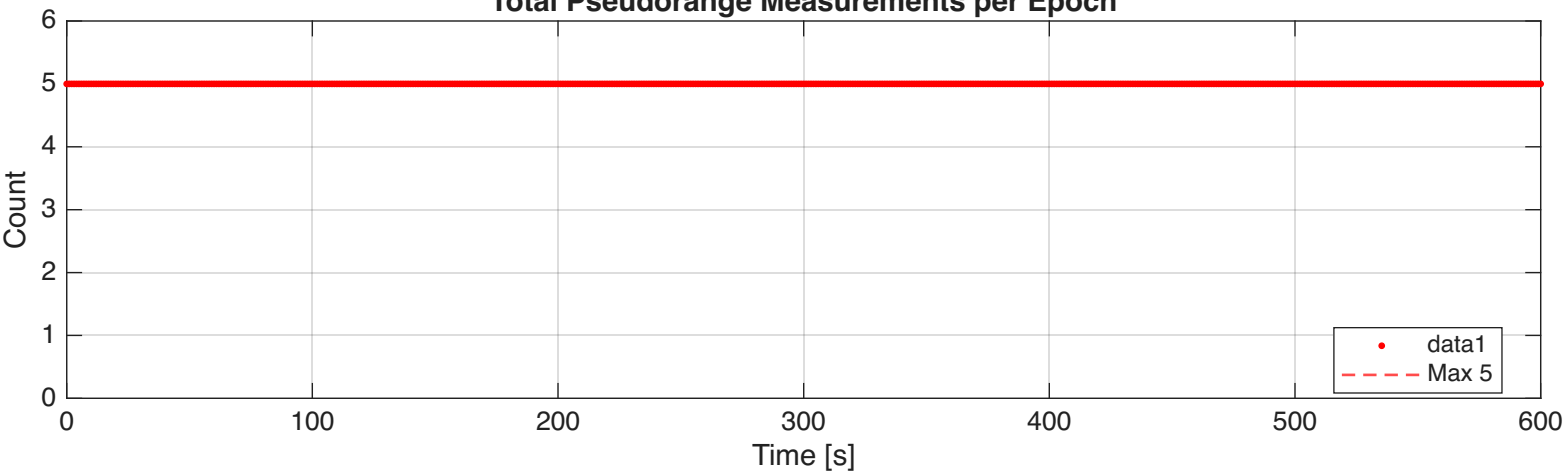


Observation Count (max meas = 5)

Visible Ground Towers per Epoch



Total Pseudorange Measurements per Epoch

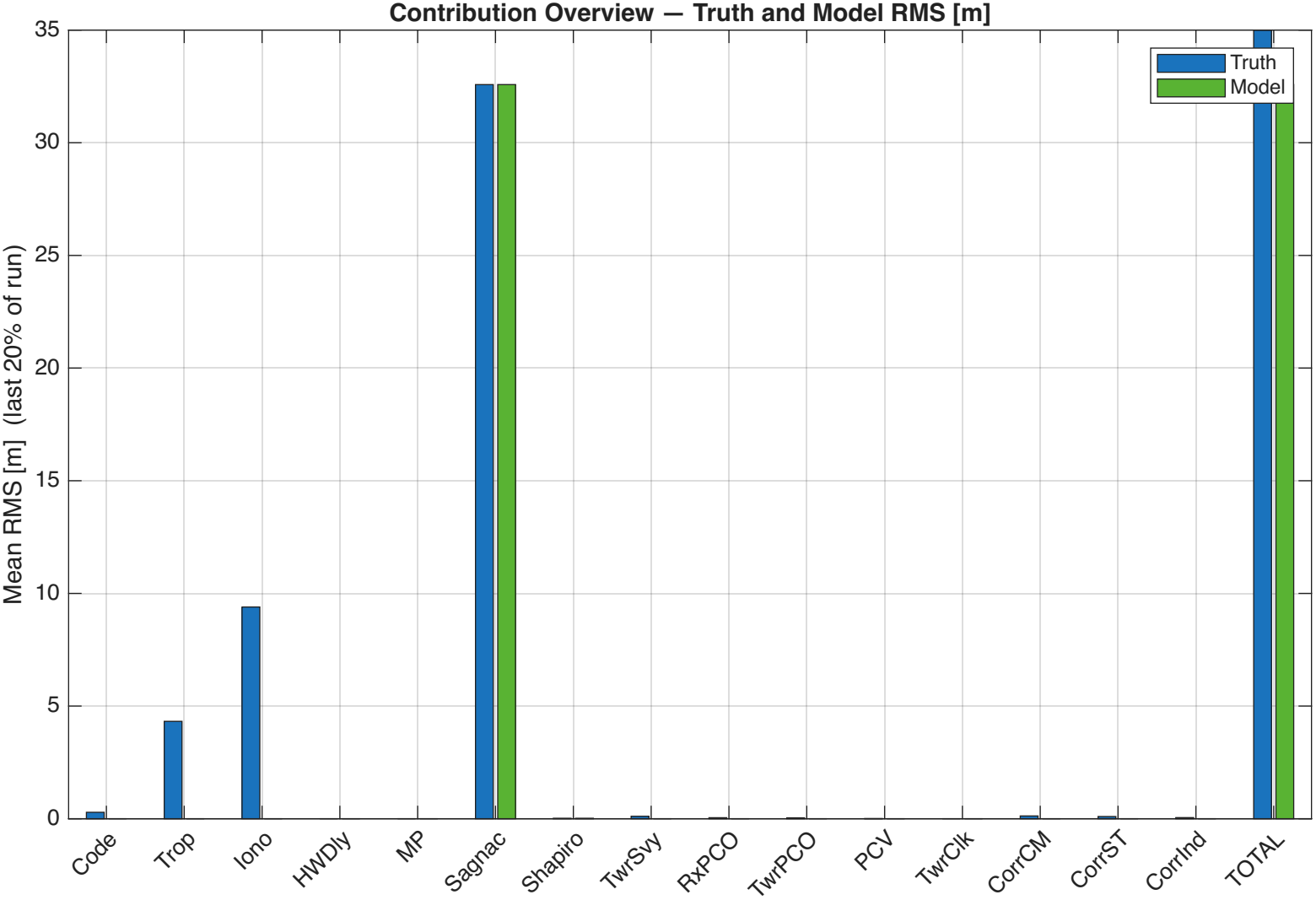


Contribution Overview — Truth and Model RMS [m]

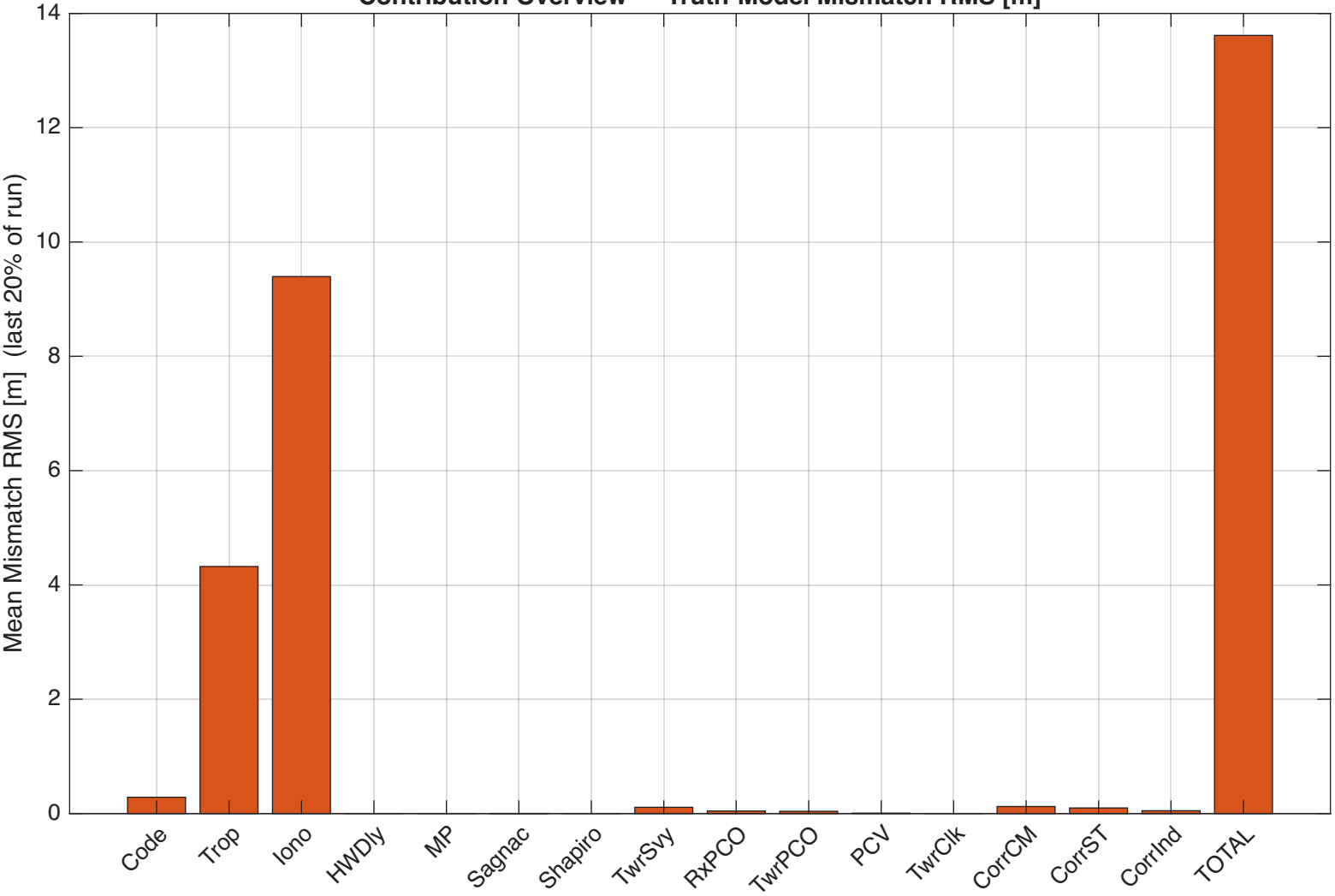
Mean RMS [m] (last 20% of run)

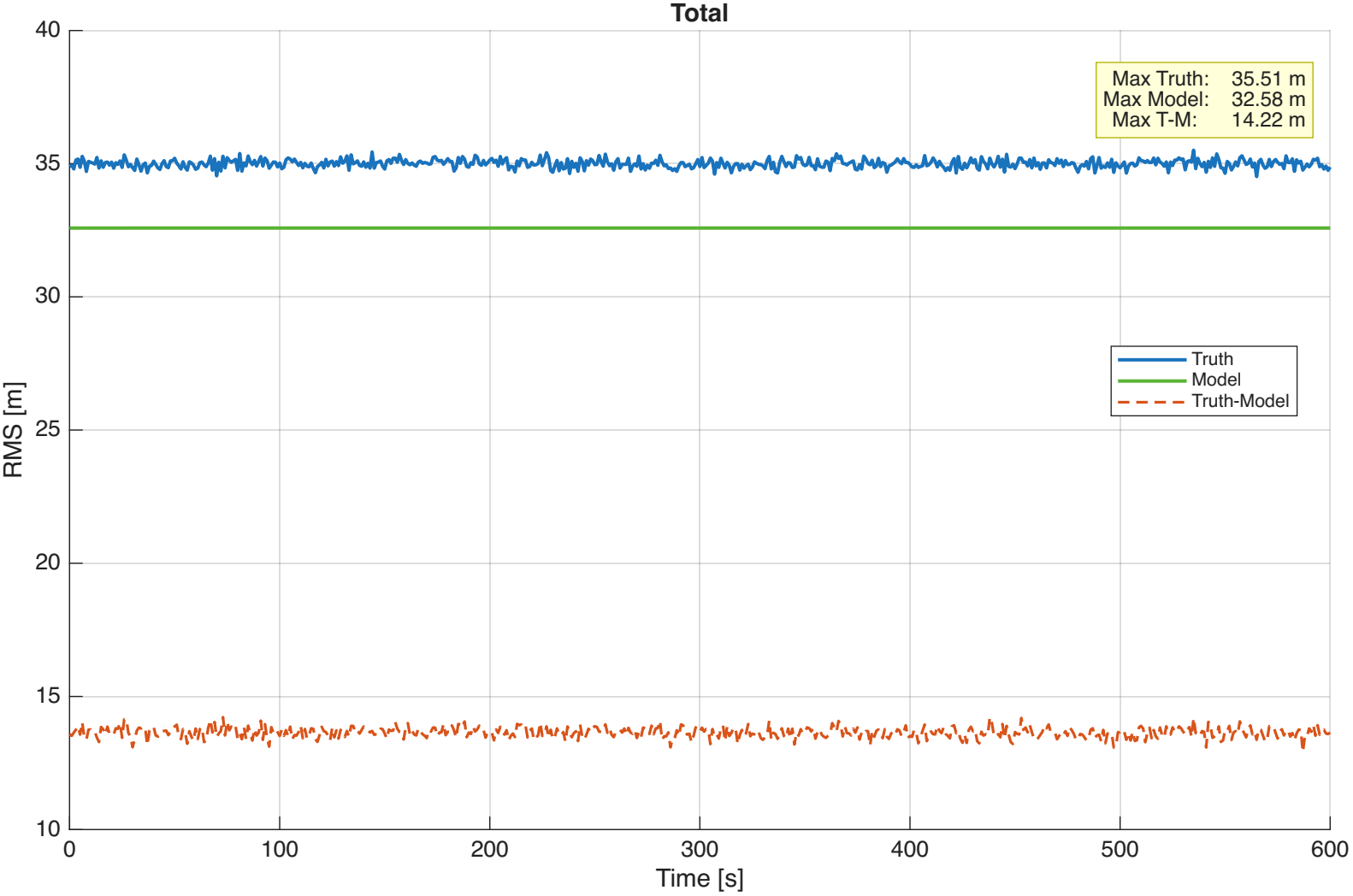


Code Trop Iono HWDly MP Sagnac Shapiro TwrSvy RxPCO TwrPCO PCV TwrCik CorrCM CorrST CorrInd TOTAL

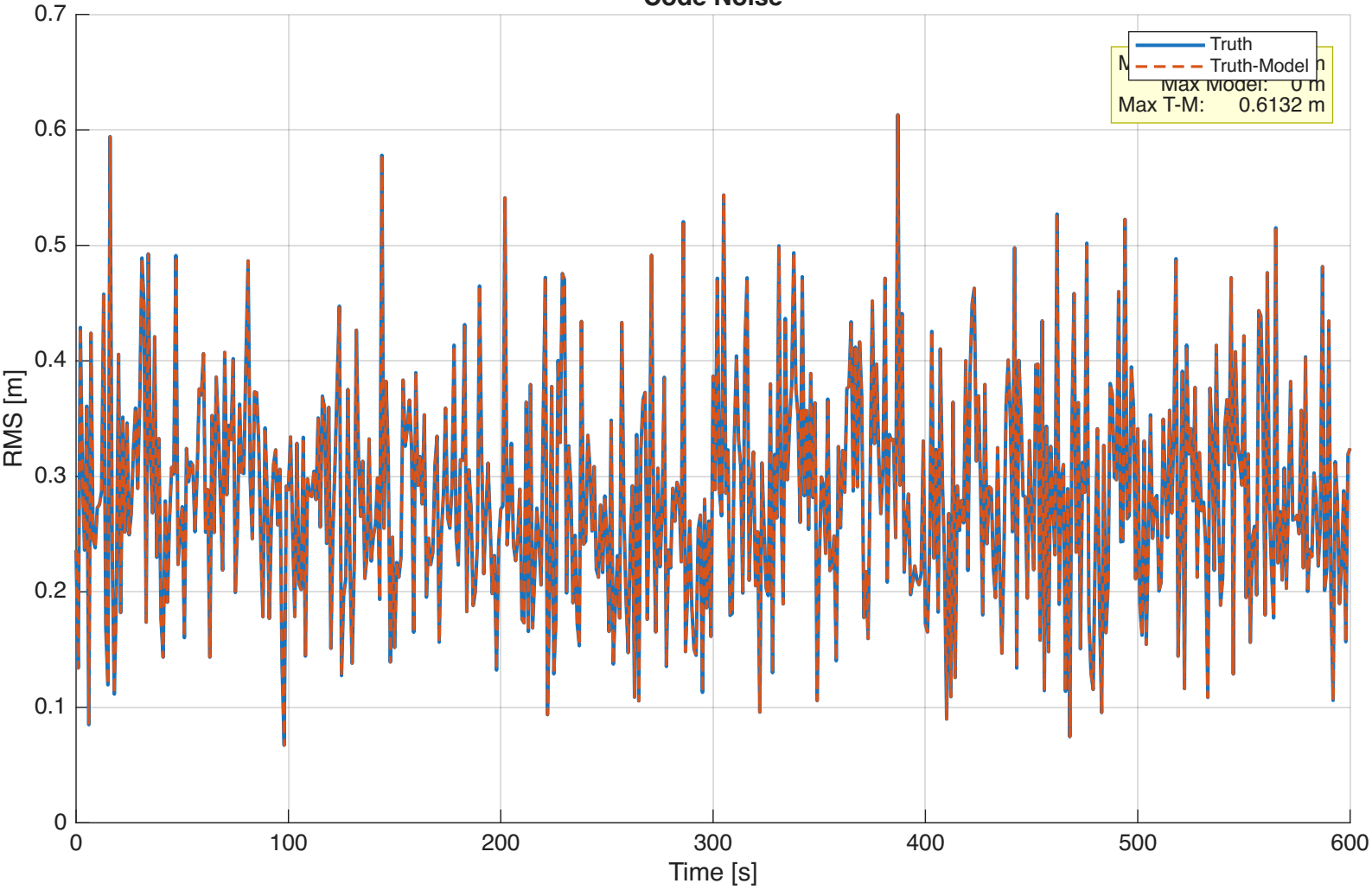


Contribution Overview — Truth-Model Mismatch RMS [m]

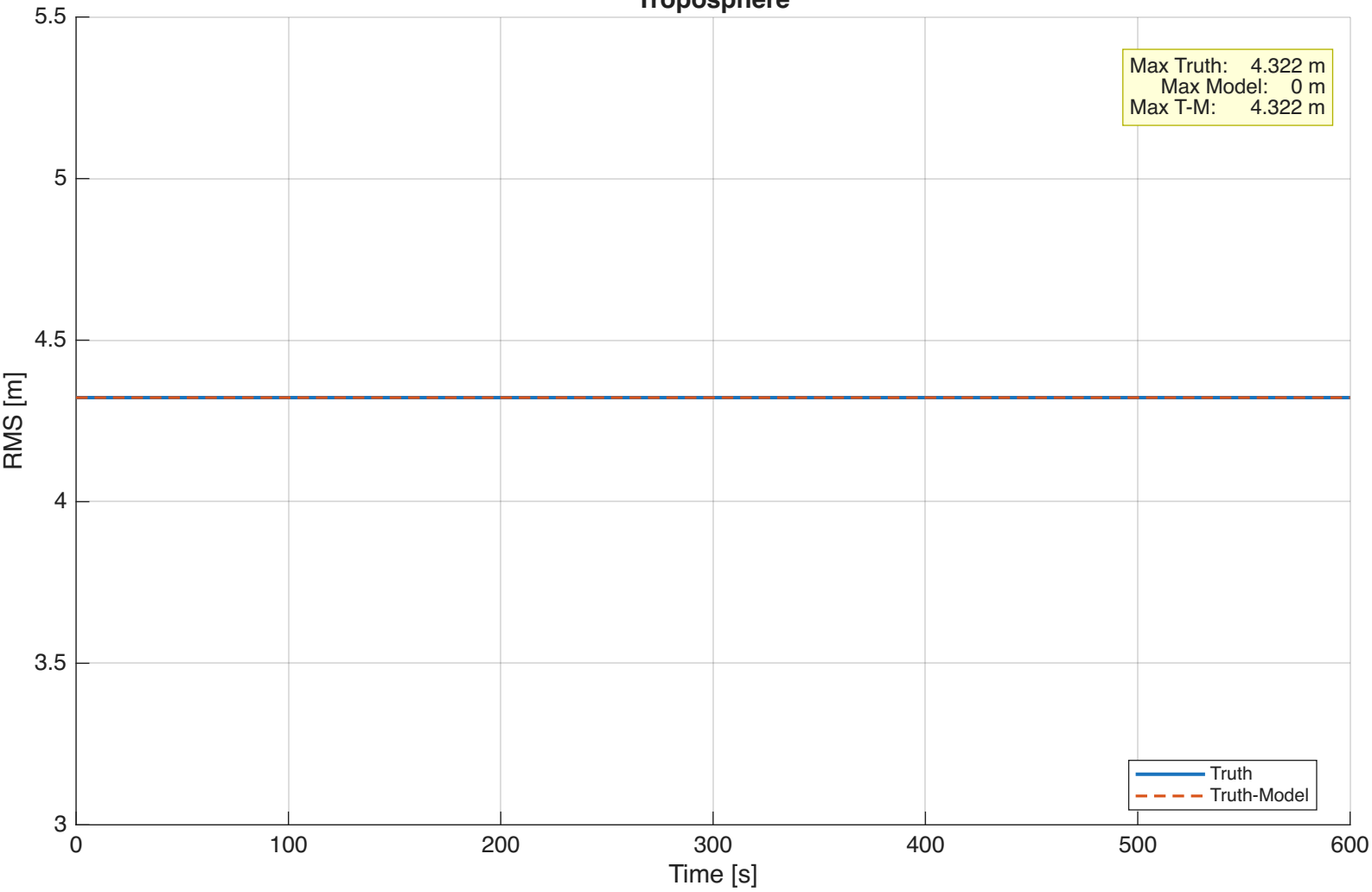




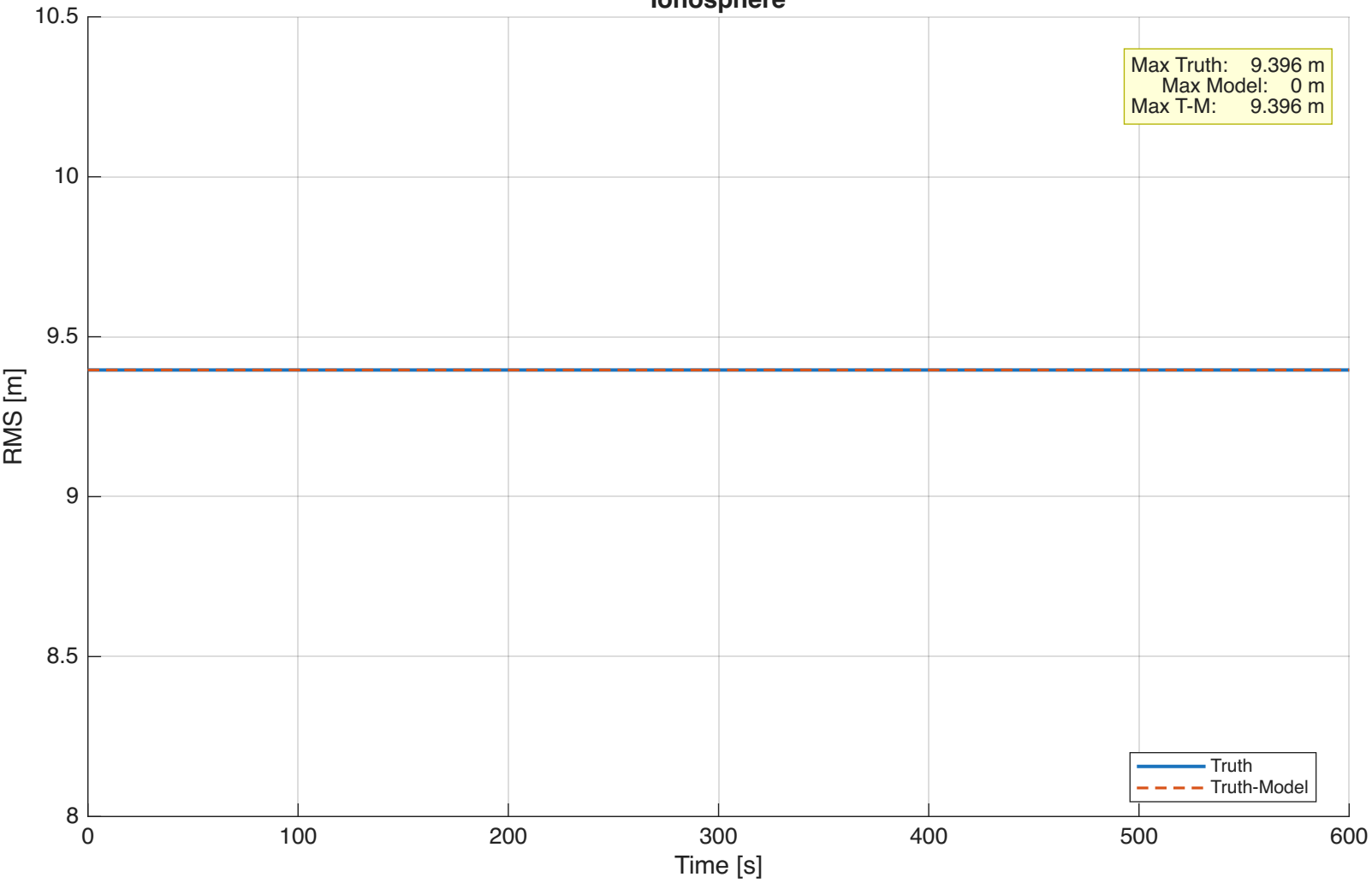
Code Noise



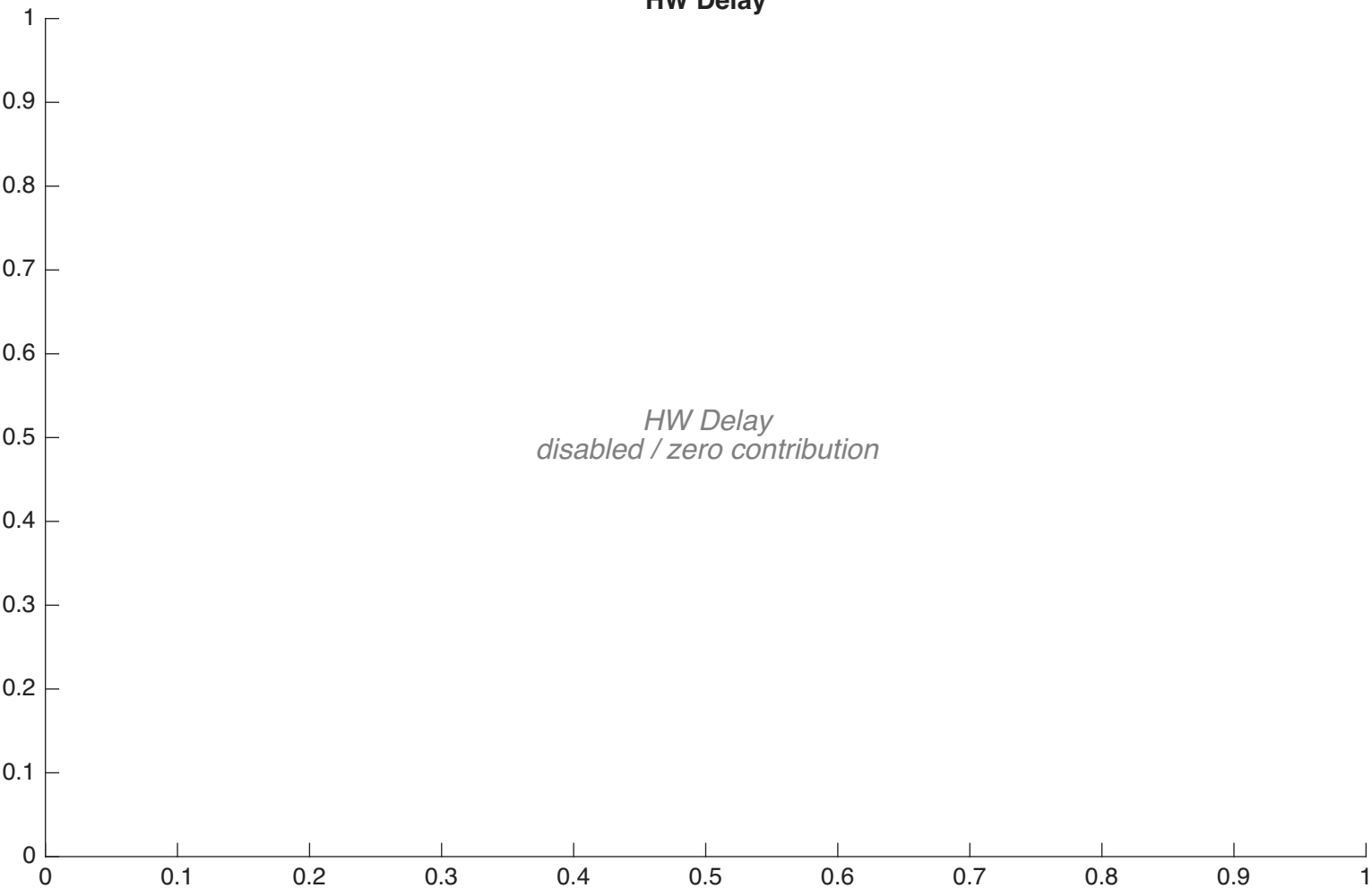
Troposphere



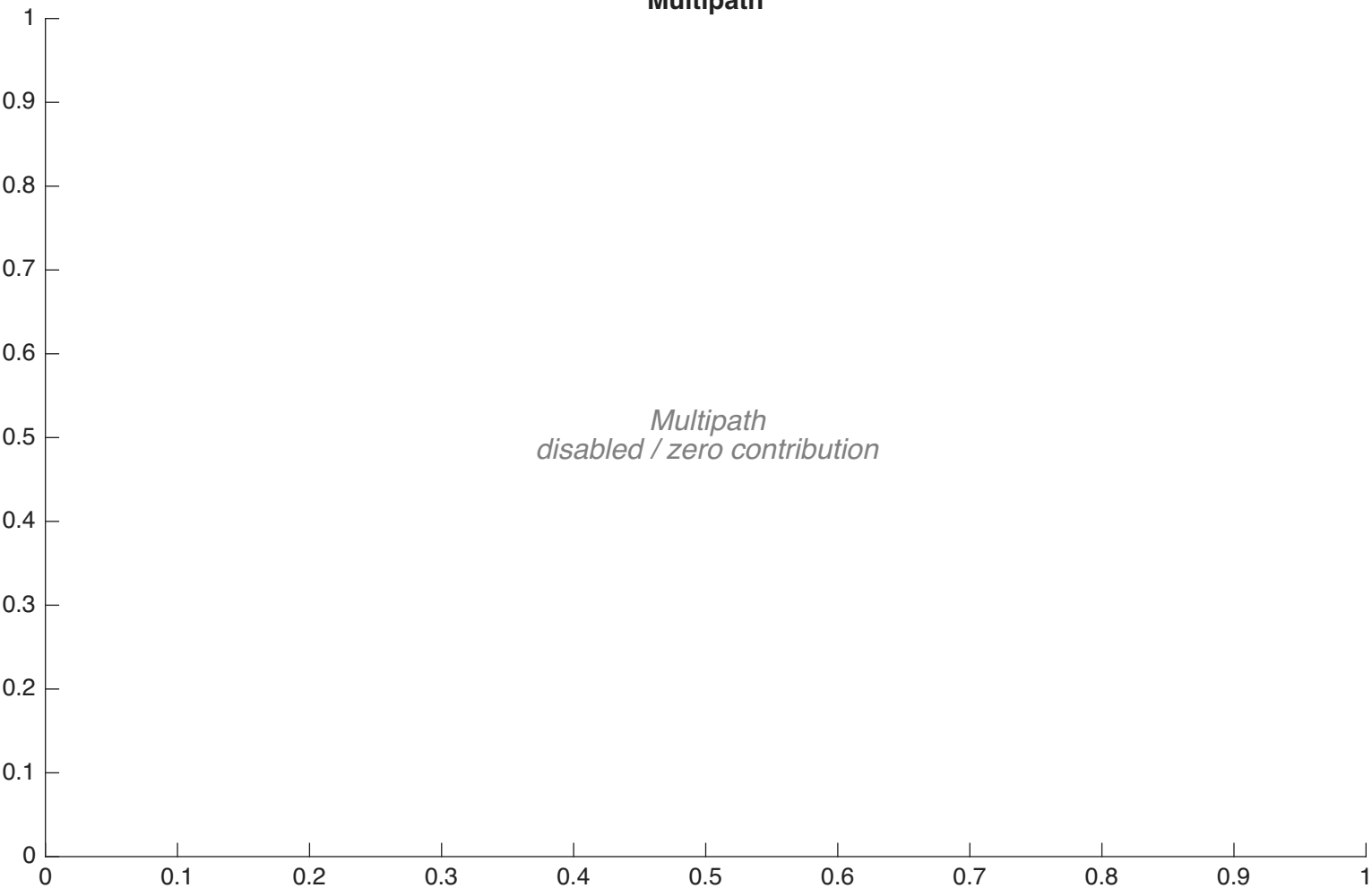
Ionosphere



HW Delay

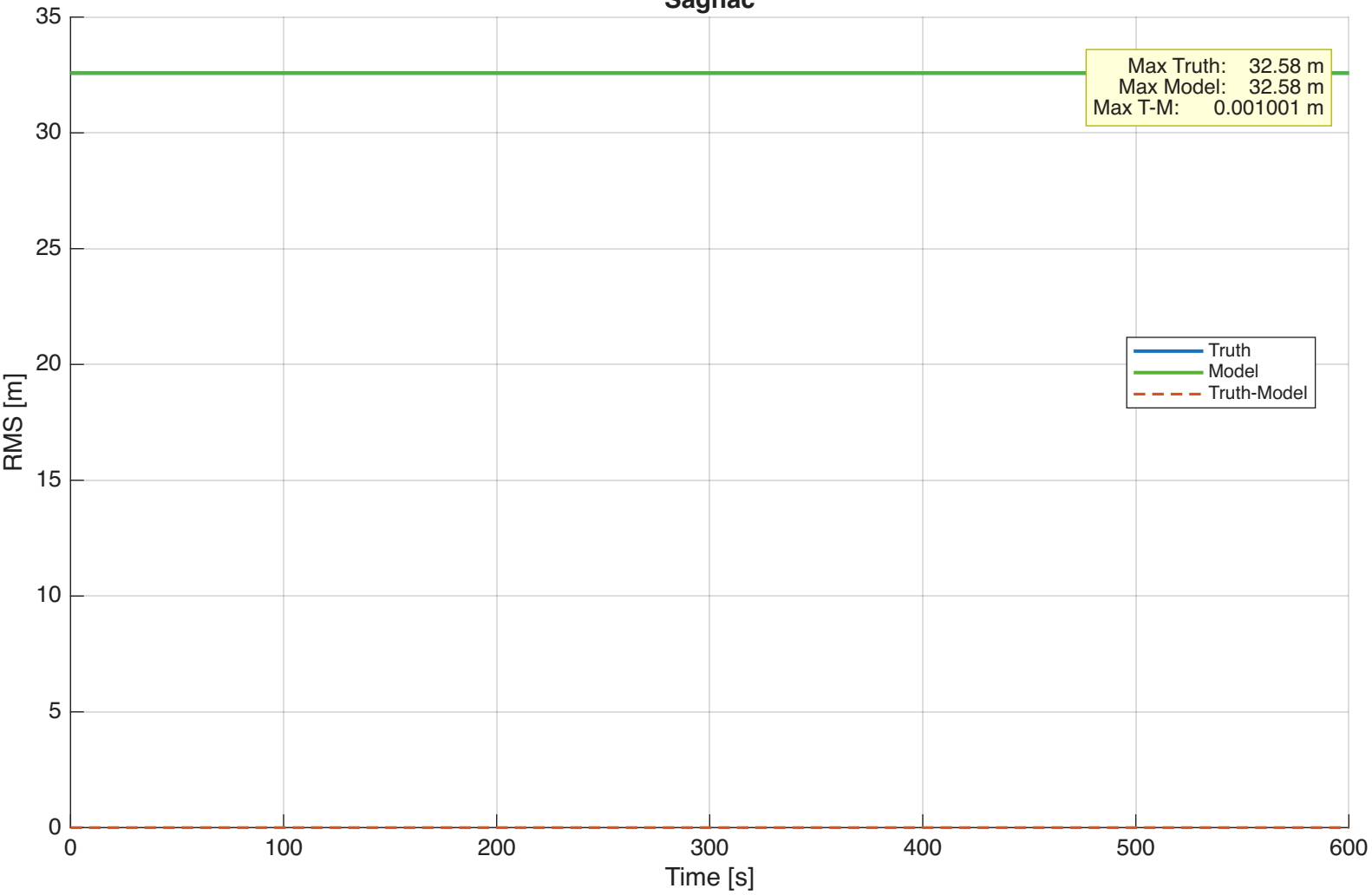


Multipath

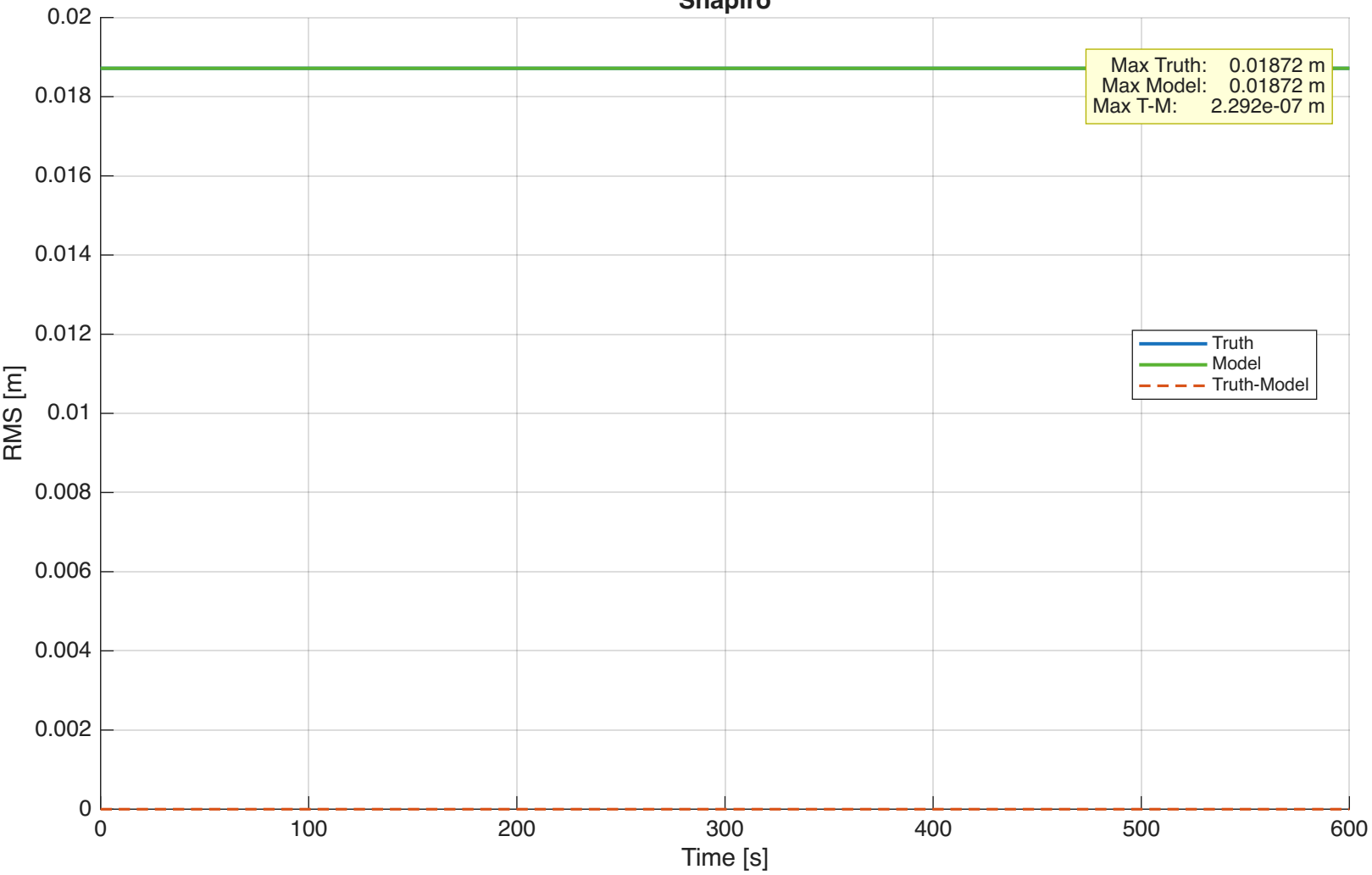


*Multipath
disabled / zero contribution*

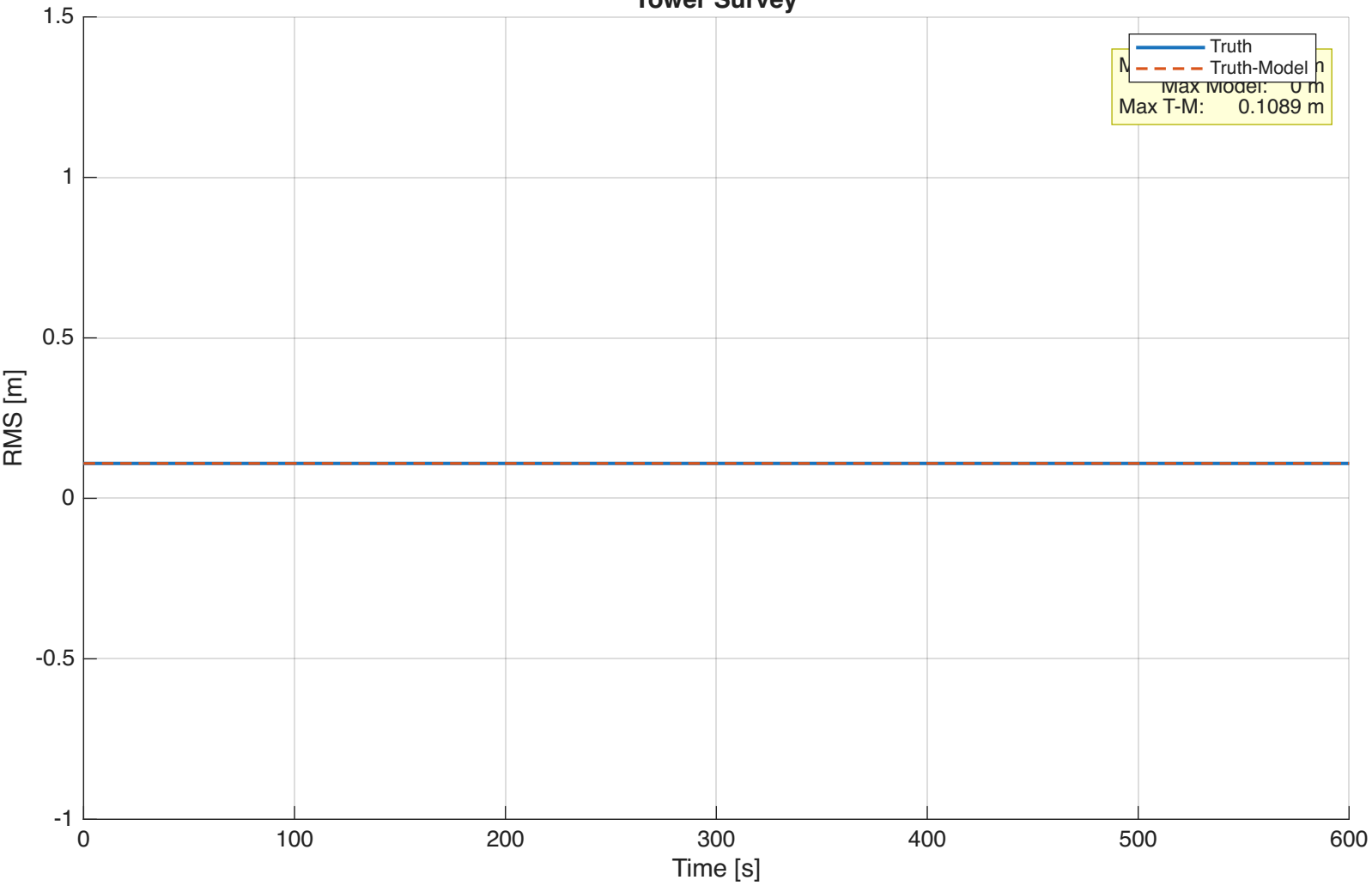
Sagnac



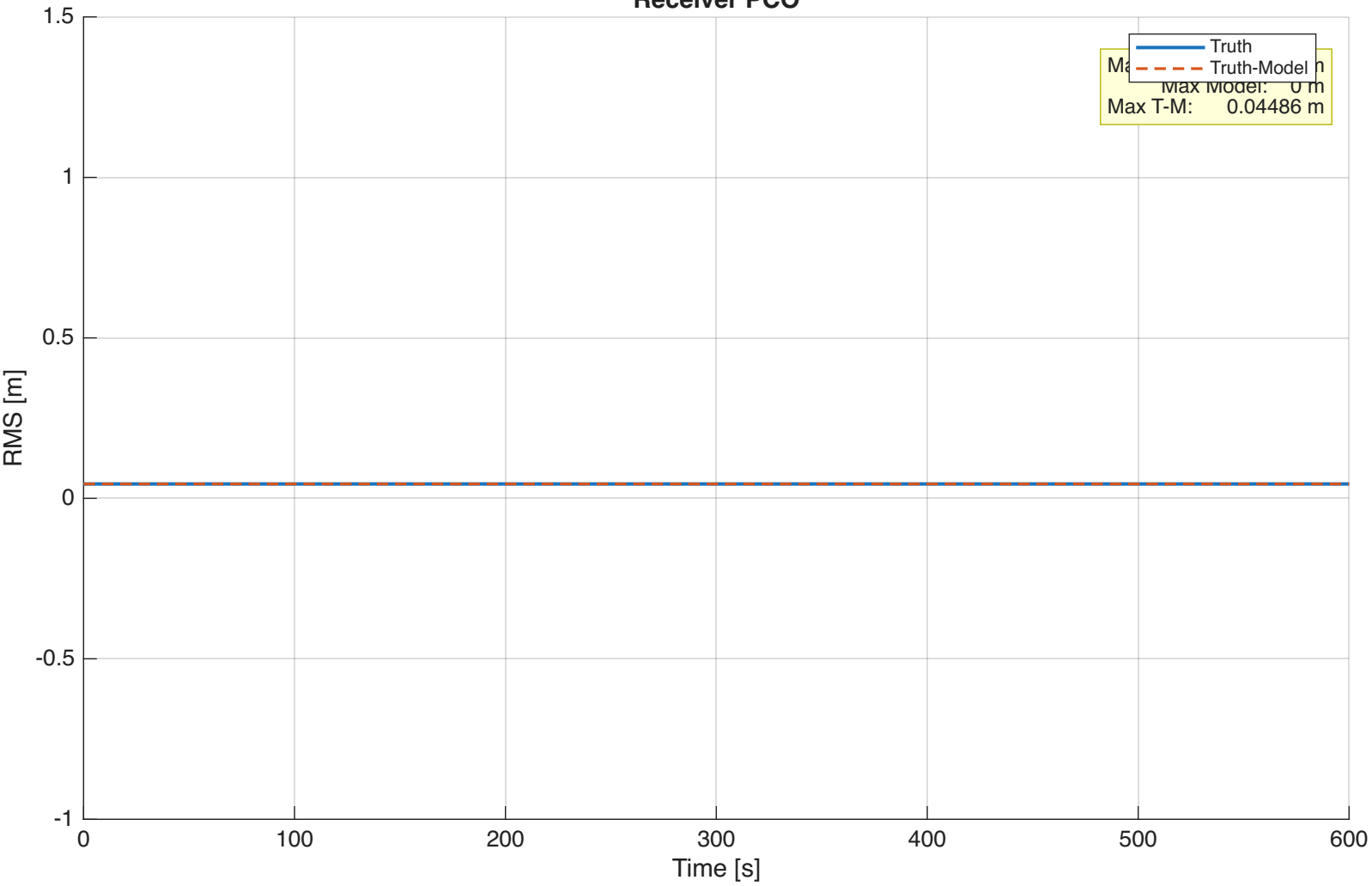
Shapiro



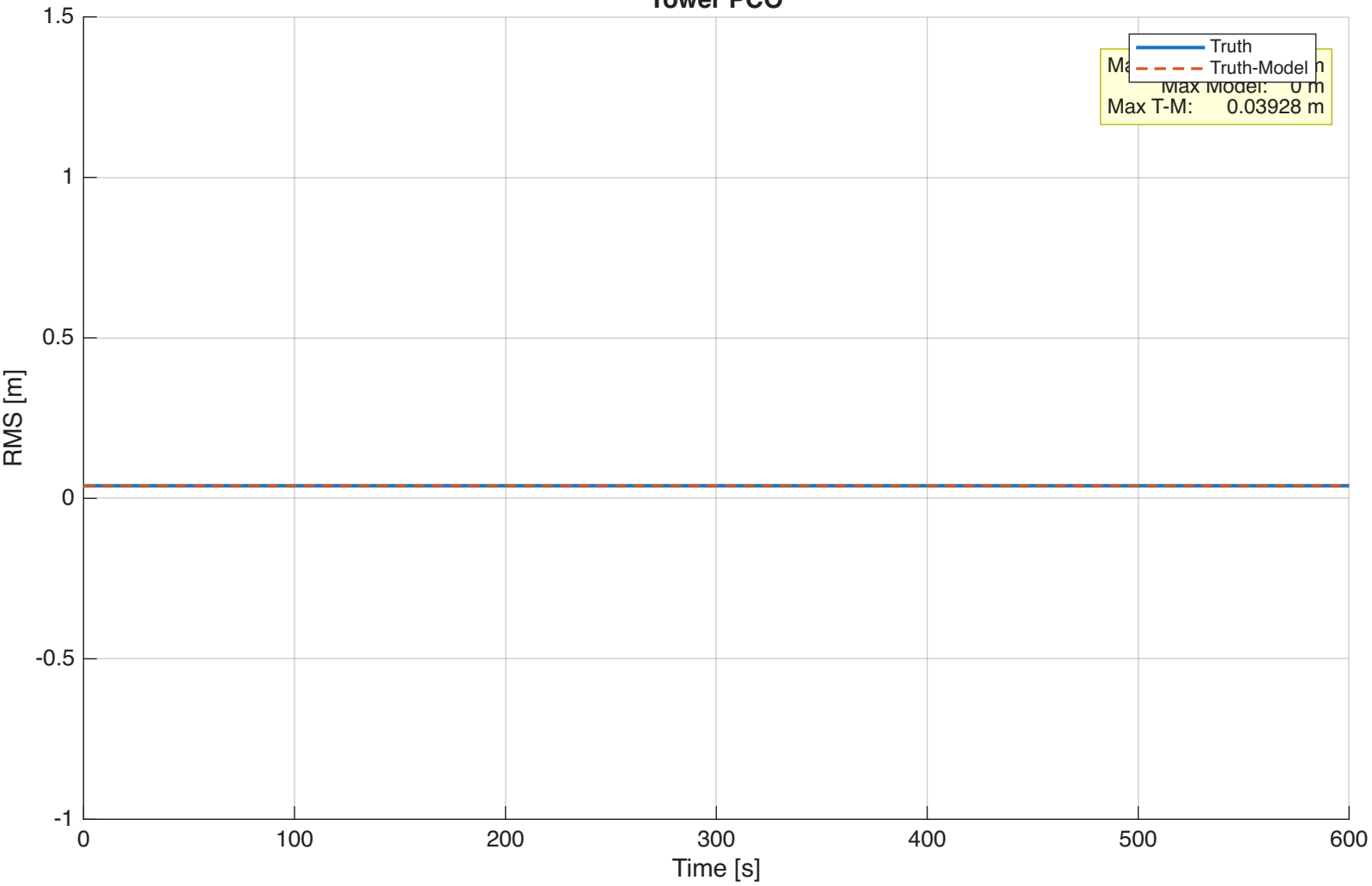
Tower Survey

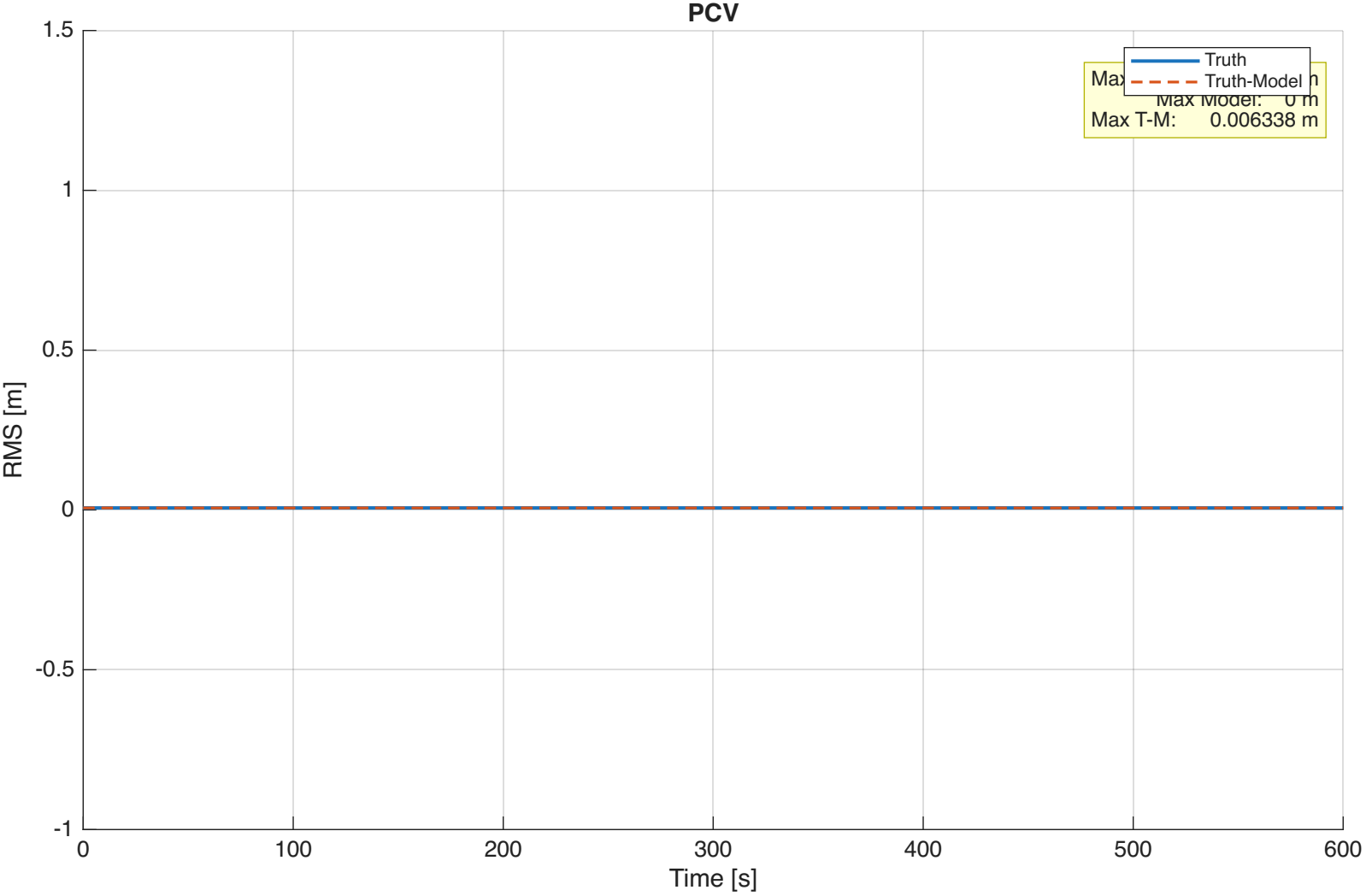


Receiver PCO



Tower PCO



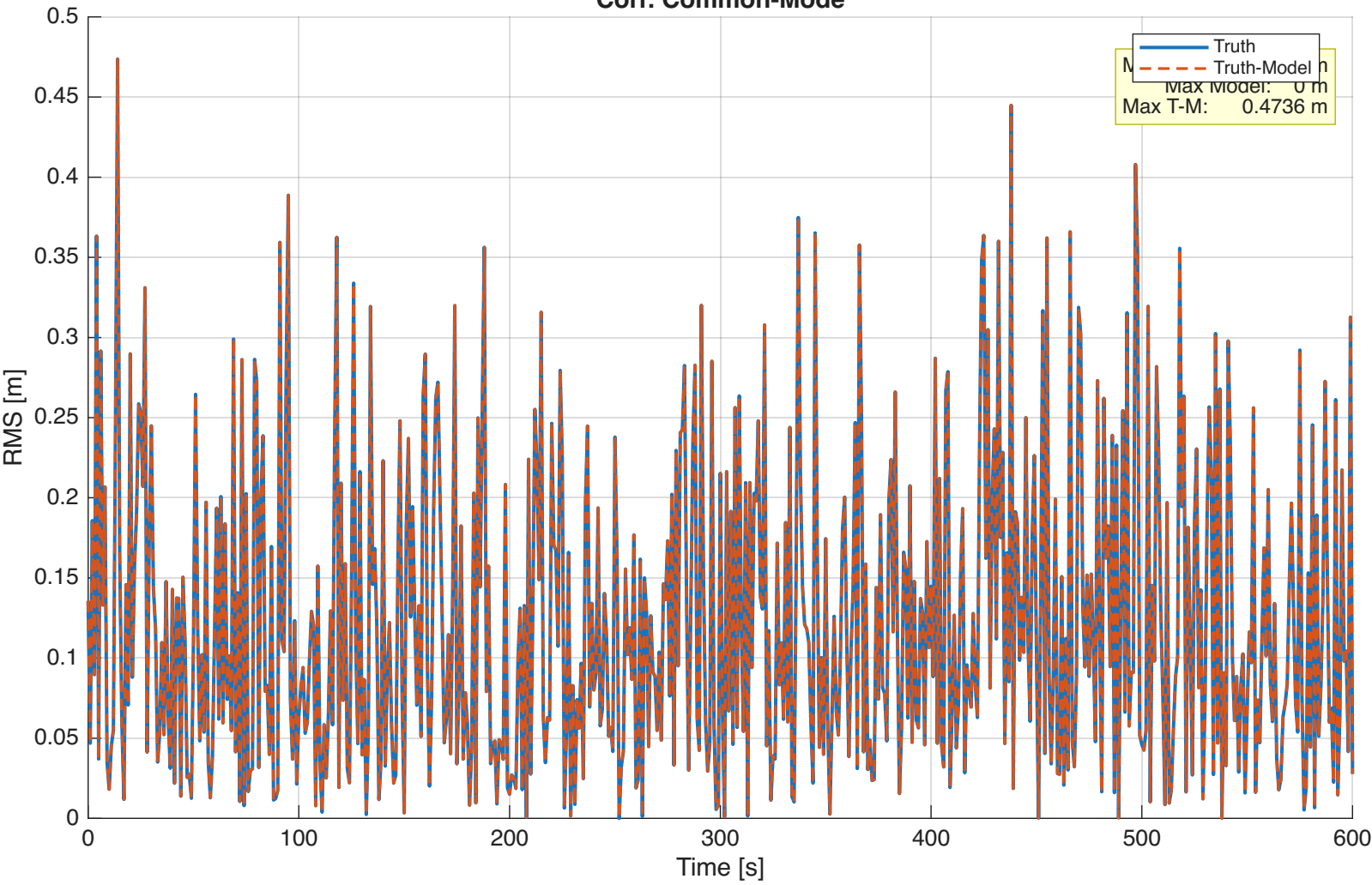


Tower Clock

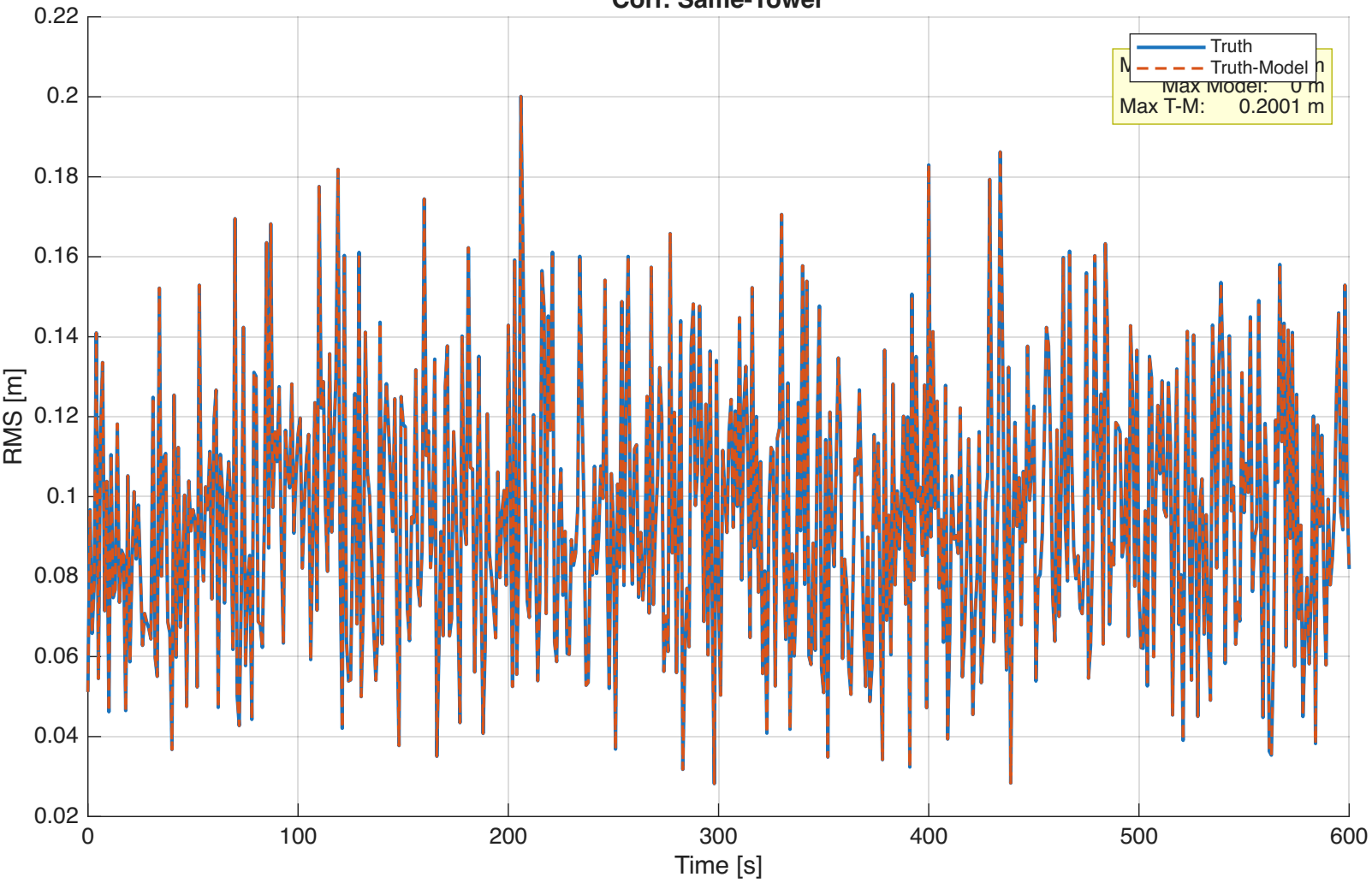
Tower Clock

ENABLED BUT NO DIAGNOSTIC DATA FOUND

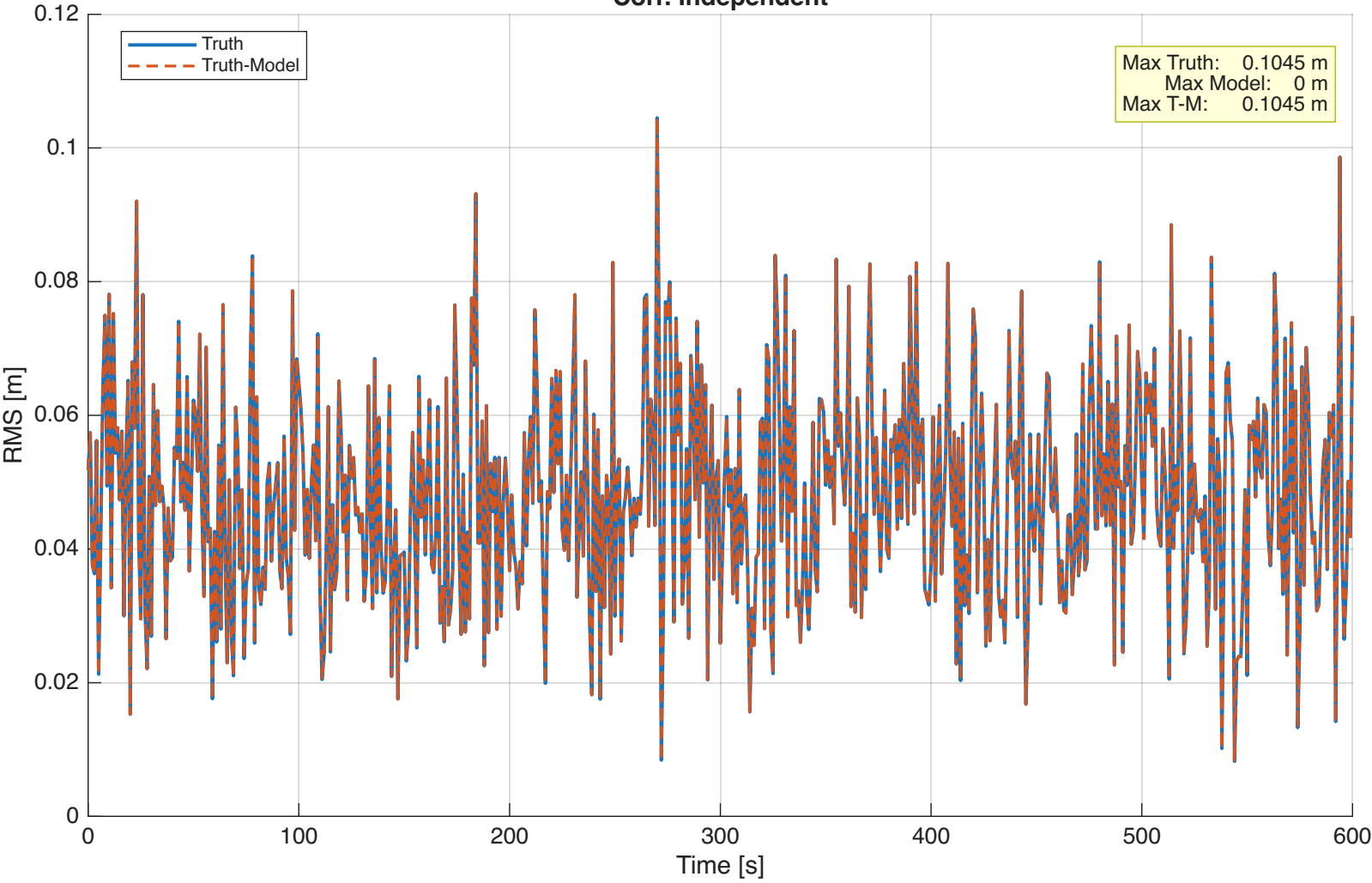
Corr. Common-Mode



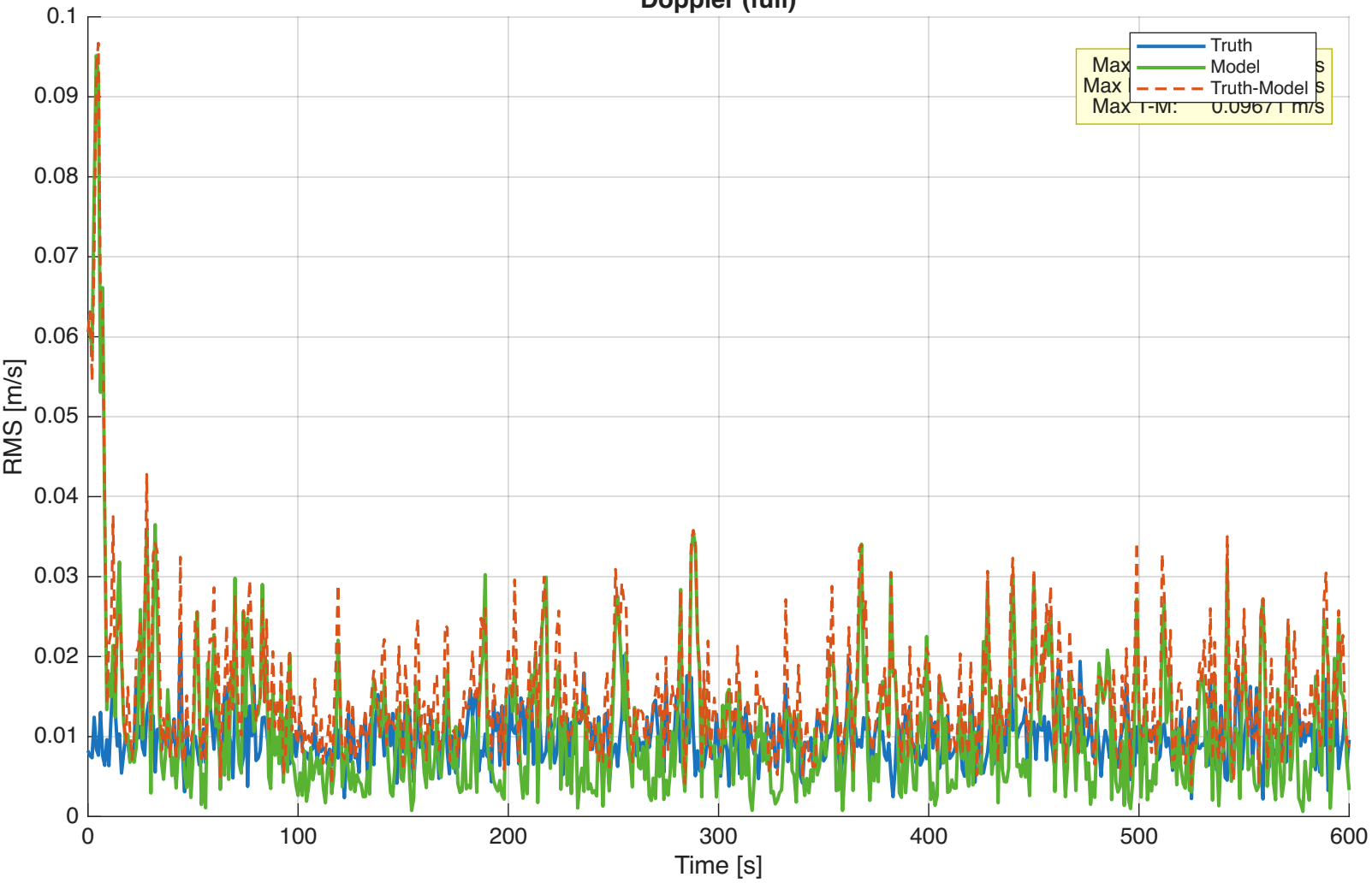
Corr. Same-Tower



Corr. Independent



Doppler (full)



Doppler Twr Clk Drift

